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INTRODUCTION

During the lifetime of a hemodialysis patient, he/she will require many vascular access procedures including frequent placement of temporary or tunneled hemodialysis catheters, arteriovenous (AV) fistula and AV graft placements and revisions, surgical or percutaneous thrombectomies of AV fistulas and grafts, and other related endovascular procedures for AV access. For many years the majority of vascular access procedures were performed by surgeons and interventional radiologists. However, in recent years nephrologists have strived to improve the quality and timely provision of these vascular access services, and have formed a new subspecialty in nephrology: interventional nephrology. As a result, dialysis access procedures are increasingly being performed by appropriately trained nephrologists. Furthermore, in recent years interventional nephrologists have expanded the services they perform and many interventional nephrologists also are trained to create primary AV fistulas and grafts, insert peritoneal dialysis catheters, and perform native kidney biopsies. Issues relating to these procedures will be discussed in this chapter.

OVERVIEW OF VASCULAR ACCESS FOR DIALYSIS

The majority of patients with end-stage kidney disease (ESKD) receive hemodialysis three times weekly to optimize their survival, minimize medical complications, and enhance quality

of life.¹ Thus, vascular access is the lifeline for a patient receiving hemodialysis. A reliable and durable vascular access is a critical requirement for provision of adequate hemodialysis. The ideal vascular access would be easy to place, ready to use as soon as it is placed, deliver high blood flows indefinitely, and be free of complications. None of the existing types of vascular access achieves this ideal. Among the three types of vascular access currently available, native AV fistulas are superior to AV grafts, which in turn are superior to dialysis catheters.

Native AV fistulas are created by performing a direct anastomosis between a high-pressure artery and a low-pressure vein. Exposure of the vein to the high arterial pressure causes it to dilate and increase its blood flow. To be used reproducibly for dialysis, an AV fistula must have a large enough diameter to be safely cannulated with large-bore dialysis needles, and a sufficiently high access blood flow to permit blood flow ≥ 350 mL/min. It also must be superficial enough for the landmarks to be easily appreciated by the dialysis staff performing the cannulation. Increases in blood flow and draining vein diameter occur fairly rapidly following AV fistula creation. Whereas blood flow in a normal radial artery is only 20 to 40 mL/min, it increases more than 10-fold within a few weeks of fistula creation. In one study, the mean access blood flow in successful AV fistulas was 634 mL/min two weeks postoperatively²; in a second study it was 650 mL/min 12 weeks following AV fistula creation.³ Moreover, the mean access blood flows and AV fistula diameters are not significantly different in the second, third, and fourth months following AV fistula creation.⁴ This implies that determination of whether a new AV fistula is likely to be used successfully for

dialysis should be possible within 4 to 6 weeks of the initial surgery. AV fistulas should not be cannulated immediately as the maturation process requires several weeks to months.

AV grafts are artificial connections between arteries and veins most commonly consisting of polytetrafluoroethylene (PTFE), which can be cannulated very early, ideally 3–4 weeks following surgical creation, once the AV artery and vein heal following the surgical procedure and the surrounding edema has settled.

Dialysis catheters include temporary large-bore venous catheters which are usually only used short-term, and tunneled, cuffed permanent venous catheters. Of the long-term access options, tunneled catheters have the highest frequency of infection and thrombosis, but are a necessary evil, either as a bridge device in patients waiting for an AV fistula to mature or an AV graft to heal, or as an access of last resort in patients who have exhausted all AV fistula or graft options.^{5,6} As compared to patients who continue to dialyze with catheters, those who switch from a catheter to an AV fistula or graft have a substantially lower mortality,^{7,8} as well as improvements in serum albumin and erythropoietin responsiveness.⁹

Recognizing the relative merits of the type of vascular access, the 2006 Kidney Disease Outcomes Quality Initiative (K/DOQI) guidelines for vascular access recommended a target use of AV fistulas in at least 40% of hemodialysis patients and catheter use in only 10% of prevalent patients.¹⁰ The proportion of U.S. patients using an AV fistula has increased progressively from 2003 to 2012 in response to the KDOQI guidelines and the Center for Medicare and Medicaid Service (CMS)'s Fistula First initiative (Fig. 68.1).¹¹ Unfortunately, there has been only a modest improvement in dialysis catheter reduction among prevalent hemodialysis patients from 27% in 2003 to 20% in 2012, well above the KDOQI target of 10%. More recently, the Fistula First initiative has raised the target goal to 66% AV fistula use in prevalent hemodialysis patients.¹¹ Distribution of vascular access types among prevalent patients in the United States is presently around 63% AV fistulas, 18% AV grafts, and 19% dialysis catheters (Fig. 68.1).¹¹ However,

despite substantial improvements in the prevalent AV fistula rates, rates in incident patients have not improved in a similar fashion. The current distribution of vascular access types among patients initiating dialysis is 20% AV fistulas, 5% AV grafts, and 75% catheters.¹¹

Patients with advanced chronic kidney disease (CKD) are regularly seen by their nephrologists and referred to different sub-specialists, including vascular surgeons and interventional radiologists, for vascular access placement for hemodialysis or peritoneal dialysis. For patients who choose hemodialysis, a vascular access plan becomes a very important aspect of care, ideally completed well before hemodialysis initiation. The goal is to have a functioning AV fistula in the large majority of patients at the time of hemodialysis initiation. Even with ongoing efforts to meet K/DOQI guideline targets, it is likely that all three types of vascular access will remain in use for the foreseeable future. Thus, substantially more clinical research will be necessary to learn how to minimize complications of, and optimize outcomes associated with, each type of vascular access, as well as basic research to elucidate the pathophysiology of vascular access dysfunction and failure to develop novel and targeted therapies.

INTERVENTIONAL NEPHROLOGY IN MAINTENANCE OF HEMODIALYSIS VASCULAR ACCESS

During the course of their vascular access history, many patients undergo frequent placement of temporary or tunneled hemodialysis catheters, revision of a permanent access, surgical or percutaneous thrombectomies, and other related endovascular procedures. For many years, nephrologists took a passive role in this critically important area of dialysis care. However, during the past 20 years nephrologists have strived to improve the quality and timely provision of these services. As a consequence, dialysis access procedures are increasingly being performed by appropriately trained nephrologists who

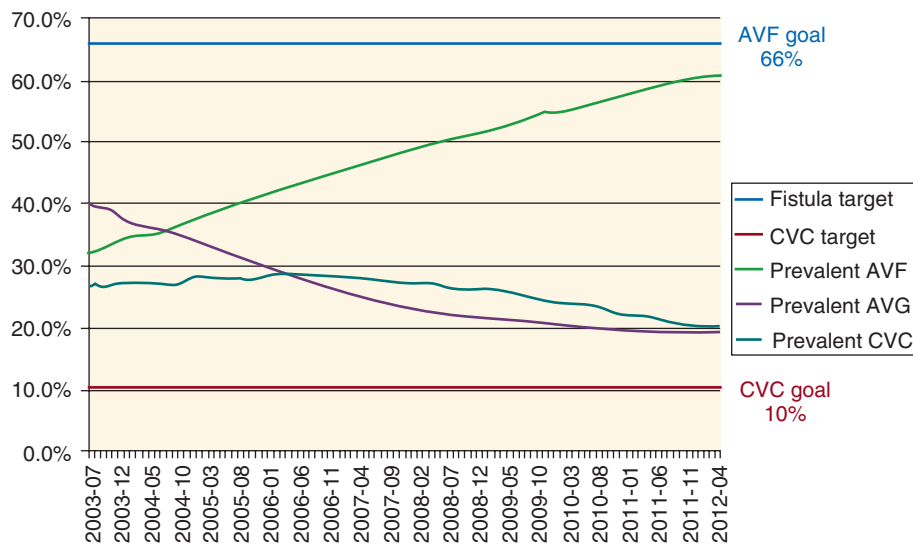


Fig. 68.1 Impact of Fistula First Breakthrough Initiative (FFBI): Since initiation of FFBI there has been an increase AV fistula use (AVF) and reduction in AV graft (AVG) use. There has been only a modest reduction in central venous catheter (CVC) use. (Adapted from Fistula First Breakthrough Initiative Dashboard, www.fistulafirst.org.)

know the patients extremely well and are focused on these procedures.¹²

In the past, interventional techniques to address problems with vascular access were underutilized and care of the dysfunctional access was primarily surgical. However, since publication of the initial KDOQI guidelines in 1997 and its subsequent updates, nephrologists (and interventional radiologists) have become directly involved in providing vascular access procedures. Interventional nephrology was pioneered by Gerald Beathard,¹³ and subsequently adopted by nephrologists at other medical centers.^{14–19}

In 2000, a group of interventional nephrologists and radiologists, under the leadership of Dr. Beathard, formed the American Society of Diagnostic and Interventional Nephrology (ASDIN).²⁰ This organization provides certification to interventional nephrologists, as well as accreditation to the institutions involved in the practice and teaching of interventional procedures in the nephrology specialty.¹⁷ Training, certification, and accreditation are given for diagnostic ultrasound, peritoneal dialysis catheter insertion, and endovascular procedures on AV fistulas and grafts and central venous catheters for hemodialysis.²¹ Comprehensive training is required to achieve dexterity and knowledge. Several academic and nonacademic centers train practicing nephrologists and nephrology fellows in the techniques and procedures pertinent to interventional nephrology. These training centers are located either in a freestanding interventional facility or in a hospital-based radiology suite. Workshops and training are increasingly being supported in low and middle income countries by nephrology organizations worldwide to improve dialysis access competence and thereby dialysis quality.

Over the past 15 years, the profile of endovascular procedures performed at freestanding vascular access centers has evolved to include more procedures in AV fistula versus AV graft, a lower proportion of AV fistula thrombectomies versus elective angioplasties, and fewer catheter placements.²² The approval of several new devices (mechanical thrombectomy devices, angioplasty balloons, hemodialysis catheters) over the past twenty years has expanded the repertoire of percutaneous interventions for vascular access.²³ A comprehensive list of procedures typically performed by interventional nephrologists is provided in Table 68.1. A given interventional

nephrology program may provide only a subset of these procedures, depending on the local needs and arrangements with radiology and surgical subspecialties. A solo nephrologist may decide to perform selected procedures necessary to provide an immediate dialysis access (such as insertion of dialysis catheters), thus eliminating delays resulting from scheduling difficulties. Other nephrology practices may include one or two nephrologists fully trained to perform a spectrum of interventional procedures.

When nephrologists perform interventional access procedures, hospitalizations for vascular access-related complications and missed outpatient hemodialysis sessions because of vascular access problems decrease.^{19,24–26} Given the increasing numbers of chronic kidney disease patients worldwide, the role of the interventional nephrologist may need to expand more consistently from vascular maintenance to include AV access creation.

At University of Alabama at Birmingham, a unique multidisciplinary model has been adopted to streamline vascular access management. This model consists of a joint interventional radiology/nephrology program, with interventional nephrologists and interventional radiologists sharing the same radiology suites, and working side by side to perform all dialysis access procedures, profiting from the existing technical, clinical, imaging, and surgical competencies at the same institution.²⁷ Recent publications from other academic medical centers have also highlighted the opportunity for interventional nephrology to serve as a model for interdisciplinary training, research, and patient care.^{28–30}

A key element of any successful interventional program, whether it involves radiologists or nephrologists, is active collaboration with nephrology and dialysis programs to track outcomes of the procedures, and implement timely quality improvement initiatives. This can be best accomplished by having dedicated vascular access coordinators who maintain prospective, computerized records of all procedures performed and follow patients up regularly.²⁷

Vascular access procedures and their subsequent complications represent a major cause of anxiety, morbidity, hospitalization, and cost for patients on maintenance hemodialysis.^{31–35} In an earlier publication from 1996, more than 20% of hospitalizations in US hemodialysis patients are vascular access-related, and the annual cost of vascular access morbidity is close to 1 billion U.S. dollars.³⁴ A more recent publication from Thamer et al. reported that Medicare paid 2.8 billion U.S. dollars for dialysis vascular access services in the United States.³⁶ Complications that occur commonly with AV fistulas and grafts are stenosis and thrombosis, and contribute to the rising costs of vascular access care. AV grafts are prone to recurrent stenosis and thrombosis, and often require multiple interventions to ensure long-term patency. AV fistulas have a much lower incidence of stenosis and thrombosis than AV grafts, and require fewer interventions.^{34,35,37}

Table 68.1 Procedures Performed by Interventional Nephrologists

Placement of non-tunneled and tunneled dialysis catheters
Exchange of tunneled dialysis catheters
Implantation of subcutaneous dialysis devices
Preoperative vascular mapping
Surveillance for stenosis
Diagnostic fistulograms of grafts and fistulas
Angioplasty of peripheral and central stenosis
Deployment of endoluminal stents for peripheral and central stenosis
Thrombectomy of grafts and fistulas
Sonographic or angiographic assessment of immature fistulas
Salvage procedures for immature fistulas
Placement of peritoneal dialysis catheters
Diagnostic renal ultrasound
Percutaneous renal biopsy

BIOLOGY AND PATHOGENESIS OF VASCULAR ACCESS STENOSIS IN AV FISTULAS AND AV GRAFTS

Analysis of the multicenter Hemodialysis Fistula Maturation (HFM) study revealed a progressive increase in AV fistula diameter and blood flow in the 6-week period following AV

fistula creation.³⁸ Incorporating data from the Dialysis Access Consortium (DAC) Fistula Trial, Dember et al. reported that AV fistula non-maturation rates in the U.S. are approximately 60% even among centers of excellence.³⁹ The magnitudes of postoperative AV fistula diameter and blood flow were directly associated with the preoperative arterial function, assessed by nitroglycerin-mediated dilation and flow-mediated dilation.⁴⁰ Recent investigations from experimental models have provided valuable insight into the downstream biology of AV fistula non-maturation.⁴¹ These studies have shown that proper balance of expression of protective factors that regulate oxidative stress and endothelial function—such as heme-oxygenase-1 (HO-1) and nitric oxide synthase (eNOS) and inhibition of chemokines and mediators that modulate inflammation and vessel wall remodeling, such as monocyte chemoattractant protein-1 and matrix metalloproteinases (MMP-2 and MMP-9)—are crucial to increasing AV fistula blood flow and vasodilation and inhibiting venous neointimal hyperplasia and stenosis after AV fistula creation.^{42–44} From clinical studies, gene polymorphisms in HO-1⁴⁵ and factor V Leiden⁴⁶ have been associated with AV fistula failure.

In recent years, the preexisting health of the vein and artery at the time of access creation and its impact on AV graft and fistula outcomes has emerged as an active area of research interest.⁴⁷ Several studies have shown that preexisting venous neointimal^{48–51} and arterial neointimal hyperplasia⁵² are present in the large majority of vessels collected at the time of new vascular access surgery. Preexisting arterial intimal hyperplasia and arterial microcalcification has been associated with early AV fistula failure.⁵² A few studies have evaluated the association of preexisting venous neointimal hyperplasia and AV fistula outcomes. A recent study by Allon et al. demonstrated that preexisting venous neointimal hyperplasia

was not associated with early postoperative AV fistula stenosis.⁵³ Another single-center study of patients undergoing a two-stage AV fistula, found no significant association between preoperative venous intimal hyperplasia and postoperative neointimal hyperplasia or clinical AV fistula survival.⁵⁴ Similarly, analysis of the HFM study observed only a weak association between the magnitude of preoperative venous intimal hyperplasia with postoperative AV fistula blood flow, stenosis, or clinical maturation.⁵⁵ This suggests the possibility that outward remodeling after AV fistula creation plays a more important role in the maturation process.

In AV fistulas, neointimal hyperplasia has been shown to be the most common histology in both early and late fistula failures^{41,56–60} (Fig. 68.2). The predominant cellular phenotype present within the neointima are myofibroblasts.^{56–60} However, in AV fistula non-maturation, inadequate vasodilation (poor outward remodeling), in addition to neointimal hyperplasia, play an important role.^{58,61,62} The most common mechanism of vascular access dysfunction in AV grafts is development of venous neointimal hyperplasia at the vein-graft anastomosis^{41,58,63} (see Fig. 68.2). Venous neointimal hyperplasia is characterized by myofibroblasts and extracellular matrix components within the neointima.⁶³ In addition, angiogenesis within the adventitia and a macrophage layer lining the perigraft region also develop.⁶³

The pathogenesis of neointimal hyperplasia after AV fistula creation or graft creation placement involves a cascade of events that are best divided into upstream and downstream events⁶¹ (Fig. 68.3). Upstream events are characterized as the initial injuries to the vascular endothelium and smooth muscle cells, which include: (1) surgical trauma at the time of AV access creation; (2) hemodynamic shear stress at the vein-artery or vein-graft anastomosis; (3) inflammatory

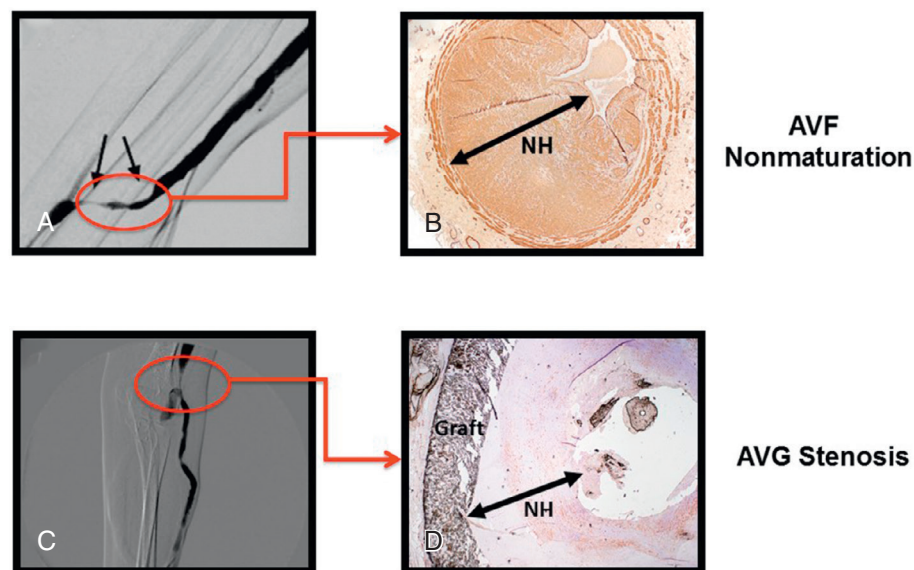


Fig. 68.2 Histologic and angiographic lesions of venous stenosis in arteriovenous fistula (AVF) and arteriovenous graft (AVG). (A) Angiographic and (B) histologic features of AVF nonmaturation. Note aggressive venous neointimal hyperplasia (NH) at vein-artery anastomosis. (C) Angiographic and (D) histologic features of AVG stenosis. Note aggressive neointimal hyperplasia at graft-vein anastomosis. Arrows show the histologic features at the site of the angiographic venous stenosis in both AVF and AVG. Images are alpha-smooth muscle actin stain and magnification is 4X. (Adapted and reprinted from Lee T, Misra S. New Insights into Dialysis Vascular Access: Molecular Targets in Arteriovenous Fistula and Arteriovenous Graft Failure and Their Potential to Improve Vascular Access Outcomes. *Clin J Am Soc Nephrol*. 2016;11(8):1504–1512, with permission from the American Society of Nephrology.)

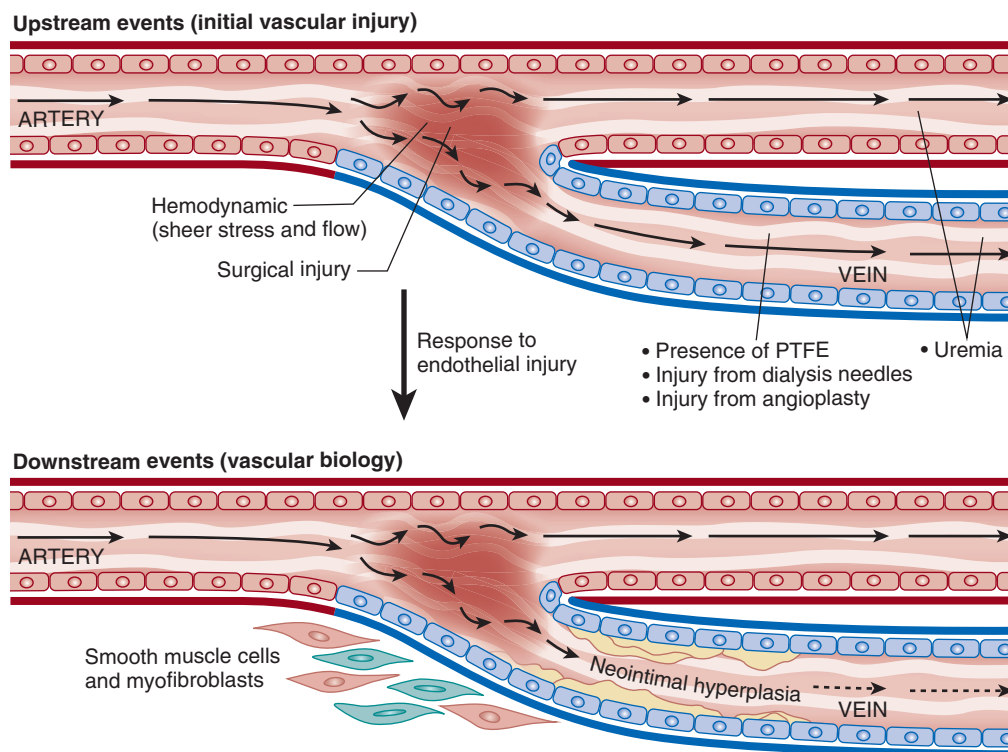


Fig. 68.3 Upstream and downstream events in hemodialysis vascular access dysfunction. Upstream events result in initial vascular injury. Downstream events are the vascular biologic response to upstream injury. Downstream biology involves mediators of oxidative stress and inflammation that regulate activation, proliferation, and migration of fibroblasts, smooth muscle cells, and myofibroblasts. *PTFE*, polytetrafluoroethylene. (Adapted and reprinted from Lee T. Novel paradigms for dialysis vascular access: downstream vascular biology—is there a final common pathway? *Clin J Am Soc Nephrol*. 2013;8(12):2194–2201, with permission from the American Society of Nephrology.)

response to the PTFE grafts; (4) injury from vessel cannulation; (5) the effects of uremia on the veins and arteries leading to endothelial dysfunction; and (6) angioplasty-related injury.^{58,59,61} Downstream events represent the vascular biological response to these upstream vascular injuries (Fig. 68.3). An inappropriate downstream response will lead to activation, proliferation, and migration of fibroblasts, smooth muscle cells, and myofibroblasts, ultimately leading to the development of neointimal hyperplasia.^{58,59,61,62} In AV fistulas, vasodilation is a necessary downstream response for maturation, in addition to inhibition of neointimal hyperplasia^{41,61,62} (Fig. 68.4).

PREOPERATIVE VASCULAR MAPPING

The need to increase primary placement of native AV fistulas has been highlighted by the K/DOQI guidelines⁶⁴ and by the Fistula First initiative (www.fistulafirst.org). There is widespread consensus among nephrologists and surgeons about the importance of maximizing functional AV fistula prevalence in patients receiving hemodialysis. Achieving this goal, however, requires overcoming a number of hurdles, including timely referral of the patient with CKD to the nephrologist and access surgeon, timely placement of an AV fistula, adequate maturation of new AV fistulas, and successful cannulation of the AV fistula for dialysis.³⁵ In the past, the surgeon's decision about the type and location of vascular

access placed was determined by a physical examination of the extremity, with and without a tourniquet. This approach was prone to substantial errors. The surgeon may not be able to adequately visualize the veins in obese patients. As a result, the surgeon might place an AV graft when an AV fistula could have been feasible. In other patients, the surgeon may decide to place a radiocephalic AV fistula after visualizing a large diameter cephalic vein at the wrist. However, an unsuspected stenosis or thrombosis in a proximal portion of that vein could doom the outcome of such an AV fistula.

The use of preoperative sonographic vascular mapping substantially increases the proportion of patients receiving an AV fistula rather than an AV graft. A prospective pilot study compared the surgeon's decision about access placement in 70 consecutive patients with CKD, before and after the results of preoperative vascular mapping were provided to the surgeon.⁶⁵ In almost one third of the patients, the surgeon changed his mind about the intended access procedure after receiving the mapping results. In most of these cases, the surgeon decided to place an AV fistula rather than an AV graft, or to place an AV fistula at a different location.⁶⁵ On the basis of these promising results, a program of routine preoperative vascular mapping was implemented. The results were dramatic: as compared to the historical control period, the proportion of patients having an AV fistula placed increased from 34% to 64%. Moreover, the proportion of patients dialyzing with an AV fistula doubled from 16% to 34%.⁶⁶ Similar increases in AV fistula placement following

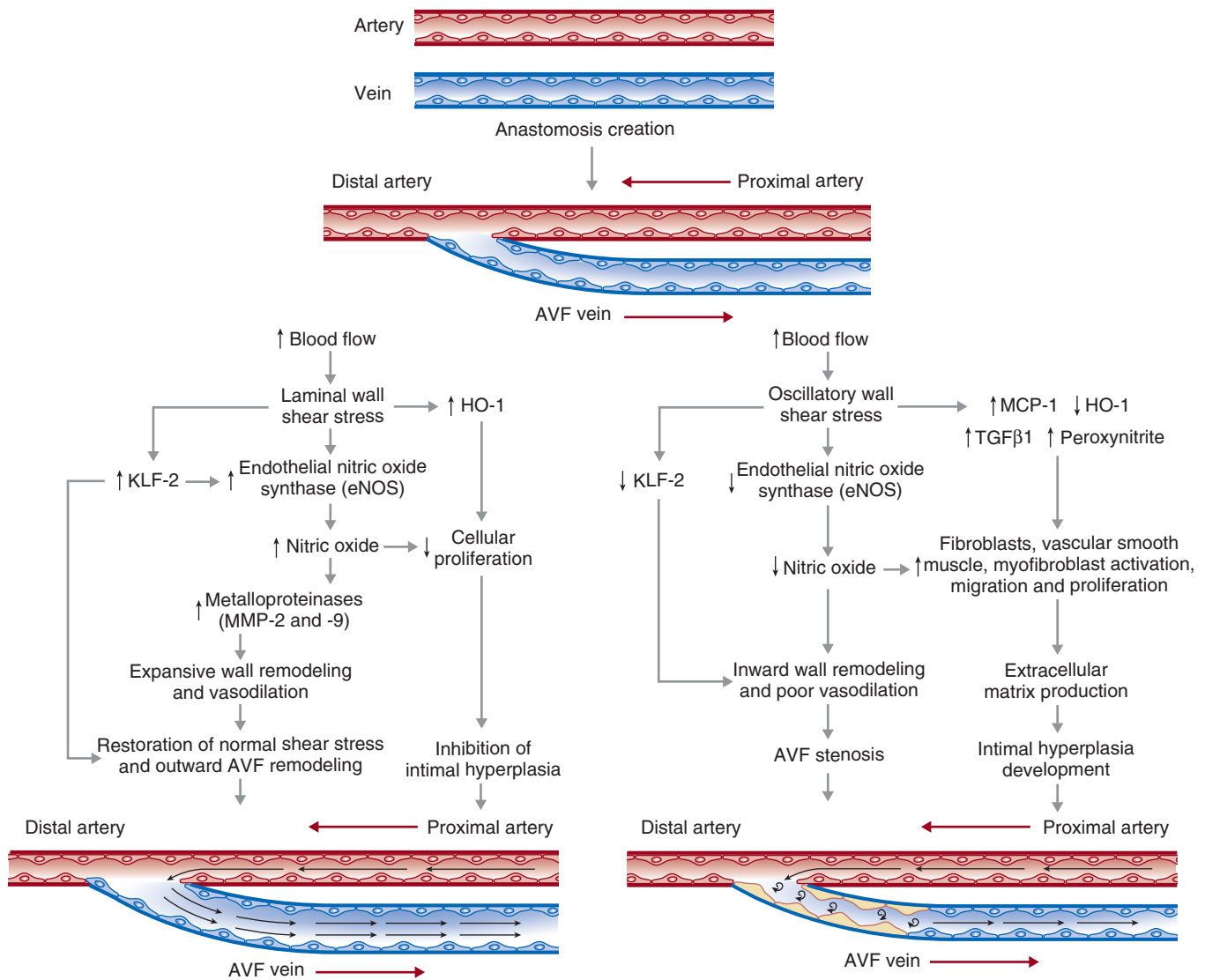


Fig. 68.4 Pathophysiologic events of successful arteriovenous fistula (AVF) maturation and AVF maturation failure. Left panel describes events of successful AVF maturation and right panel describes events of AVF maturation failure. Successful AVF maturation is dependent on outward vascular remodeling and inhibition of neointimal hyperplasia, regulated through nitric oxide production and appropriate regulation of matrix metalloproteinases. Fibroblast, smooth muscle cell, and myofibroblast activation, migration, and proliferation play a key role in neointimal hyperplasia development and AVF maturation failure. Mediators such as heme-oxygenase-1 (HO-1), monocyte chemoattractant protein (MCP-1), kruppel-like factor-2 (KLF-2), transforming growth factor beta (TGFβ1), and high levels of local oxidant stress (e.g., peroxynitrite), play essential roles in regulating cellular proliferation and neointimal hyperplasia development. (Adapted and reprinted from Lee T, Misra S. New Insights into Dialysis Vascular Access: Molecular Targets in Arteriovenous Fistula and Arteriovenous Graft Failure and Their Potential to Improve Vascular Access Outcomes. *Clin J Am Soc Nephrol*. 2016;11(8):1504–1512, with permission from the American Society of Nephrology.)

introduction of preoperative vascular mapping have been documented by other investigators (Table 68.2),⁶⁷ although a reduction in primary AV fistula failure has not been a consistent finding in all studies.^{66,68–74} Two randomized clinical trials have compared outcomes of new AV fistulas placed after preoperative ultrasound vascular mapping, as compared to those placed after clinical evaluation alone. In a Turkish study of 70 patients, the two treatment groups had similar primary and secondary AV fistula survival.⁷⁵ In a British study of 218 patients, the patients undergoing a preoperative ultrasound had a lower incidence of immediate AV fistula

failure (4% vs 11%, $p = 0.03$).⁷⁶ Primary (intervention-free) AV fistula survival was not different between the two groups (65% vs 56%, $p = 0.08$),⁷⁶ but the assisted primary (thrombosis-free) AV fistula survival was superior in the group undergoing preoperative ultrasound mapping (80% vs 65%, $p = 0.01$).

Although most centers have utilized sonographic vascular mapping, some have employed conventional venography.⁷⁷ In patients with stage 4 CKD, who have not yet started dialysis, there is a theoretical concern of radiocontrast-associated nephropathy precipitating the need for initiation of dialysis. However, in a series of 25 patients with a mean glomerular

Table 68.2 Effect of Preoperative Vascular Mapping on Vascular Access Outcomes

Reference	% Fistulas Placed		% Primary Fistula Failure		% Prevalence of Fistula Use	
	Pre-VM	Post-VM	Pre-VM	Post-VM	Pre-VM	Post-VM
Silva, 1998 ⁷⁰	14	63	36	8	8	64
Ascher, 2000 ⁶⁸	0	100	N/A	18	5	68
Gibson, 2001 ⁶⁹	11	95	18	25	N/A	N/A
Allon, 2001 ⁶⁶	34	64	54	46	16	34
Sedlacek, 2001 ⁷²	N/A	62	N/A	25	N/A	N/A
Mihmanli, 2001 ⁷³			25	6		
Miller, 1997 ⁷⁴	N/A	76				
Kakkos, 2011 ⁷¹	12	53	32	18		

N/A, Not available; VM, preoperative vascular mapping.
Adapted from Allon M, Robbin ML. Increasing arteriovenous fistulas in hemodialysis patients: problems and solutions. *Kidney Int.* 2002;62:1109–24.

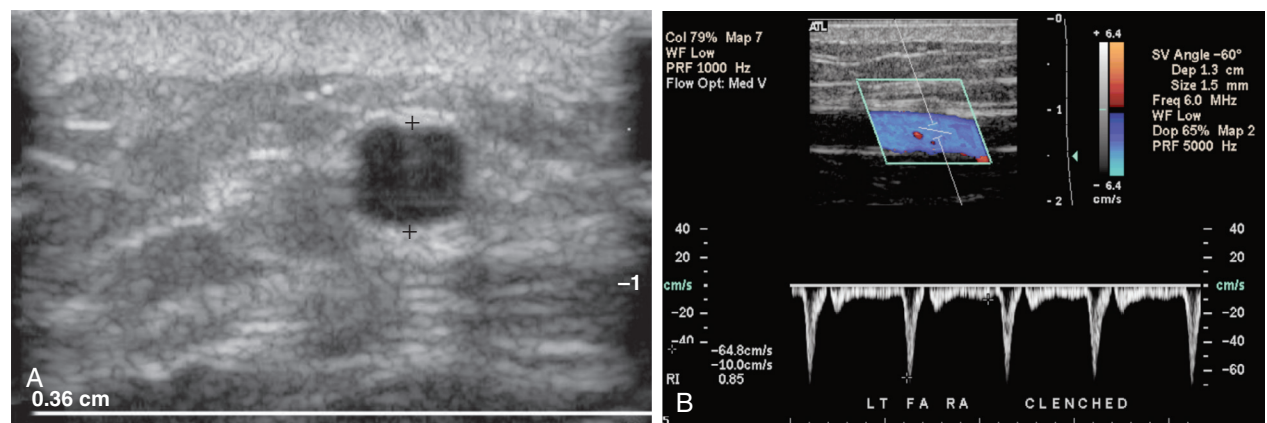


Fig. 68.5 A, Venous mapping: Venous ultrasound showing a transverse section of a cephalic vein being measured (0.36 cm). B, Venous mapping: Doppler flow measurements and color Doppler ultrasound showing a longitudinal section of a left forearm radial artery that is 1.3 cm deep and 1.5 mm in diameter.

filtration rate of 13 mL/min, none developed acute kidney injury after undergoing angiography with 10 to 20 mL of low osmolality contrast material.⁷⁸ Another potential risk of venography is injury to the very vein required for future AV fistula creation. One advantage of venography over ultrasound is its superior ability to image the central vessels and exclude central vein stenosis. Venipuncture should be performed in the hand veins if at all possible, and the cephalic vein avoided. Prospective studies are needed to further define which preoperative vascular mapping techniques are most useful in optimizing fistula maturation and in which specific situations.

TECHNICAL PROCEDURE: SONOGRAPHIC PREOPERATIVE VASCULAR MAPPING

Vascular measurements are performed with the patient in a seated position, with the arm resting comfortably on a Mayo stand. All measurements are performed in the anteroposterior dimension in the transverse plane (Fig. 68.5A, 68.5B). The minimum vein diameter for a native AV fistula is 2.5 mm.⁷⁹ The minimum vein diameter for AV graft placement is 4.0 mm.⁷⁹ The minimum arterial diameter for either AV fistula

or graft placement is 2.0 mm.⁷⁹ Veins are assessed for stenosis, thrombus, and sclerosis (thickened walls).

First, the radial artery diameter at the wrist is measured. A tourniquet is then placed at the mid to upper forearm. The veins above the wrist are percussed for two minutes, with special emphasis on the cephalic vein area. Sequential measurements are made of the cephalic vein at the wrist, and mid and cranial forearm. Any other dorsal or volar veins at the wrist are also measured and followed up the arm, according to established diameter criteria. The tourniquet is sequentially moved up the arm, and cephalic, basilic, and brachial vein diameters are measured.

After the tourniquet is removed, the subclavian and jugular veins are assessed for stenosis and thrombus. Evidence of a more central stenosis is determined by analysis of the spectral Doppler waveform for respiratory phasicity and transmitted cardiac pulsatility. If there is a clinical or sonographic suspicion of central vein stenosis, a venogram or magnetic resonance venography (MRV) is obtained.

Measurements are recorded on a worksheet. The sonographic measurements are used by the surgeon to select the most appropriate vascular access, on the basis of the following

list agreed upon by nephrologists, radiologists, and vascular surgeons, from most desirable to least desirable⁷⁹:

1. Nondominant forearm cephalic vein AV fistula.
2. Dominant forearm cephalic vein AV fistula.
3. Nondominant, or dominant upper arm cephalic vein AV fistula.
4. Nondominant or dominant upper arm basilic vein transposition AV fistula.
5. Forearm loop AV graft.
6. Upper arm straight AV graft
7. Upper arm loop AV graft (axillary artery to axillary vein)
8. Thigh AV graft

PROCEDURES INVOLVING AV GRAFTS

SURVEILLANCE FOR GRAFT STENOSIS

About 80% of AV graft failures are due to thrombosis, whereas 20% are due to infection.^{80,81} Thus, improving AV graft longevity requires implementing measures to reduce the frequency of AV graft thrombosis. Reports from United States Renal Data System (USRDS) showed that the number of angioplasties and thrombectomies (“declots”) for AV grafts had substantially increased from 0.49 to 1.10 (rate per patient year) and 0.15 to 0.48, respectively, from 1998 to 2007.¹ Among AV grafts referred for thrombectomy, a significant proportion have underlying stenosis, most commonly at the venous anastomosis, the recipient/draining vein, or the central veins.^{82–84} This observation suggests that prophylactic angioplasty of hemodynamically significant AV graft stenosis may reduce the frequency of AV graft thrombosis, and thereby increase cumulative AV graft survival.

A seminal study by Schwab et al. was the first to provide evidence supporting this approach.⁸⁵ This group of investigators measured dynamic venous pressures in AV grafts during consecutive hemodialysis sessions under carefully standardized conditions. A persistent elevation in the venous pressure measured at a low dialysis blood flow (200 mL/min) was associated with hemodynamically significant stenosis. The investigators then instituted a program of clinical monitoring for AV graft stenosis, with referral for prophylactic angioplasty if there was a suspicion of AV graft stenosis. As compared to the historical control period, a regimen of stenosis monitoring and prophylactic angioplasty reduced the frequency of AV graft thrombosis by about two thirds, from approximately 0.6 to 0.2 events per year. This landmark study stimulated a large volume of subsequent clinical research directed at two fundamental issues: (1) identifying a variety of noninvasive methods to screen for AV graft stenosis, and (2) evaluating whether stenosis surveillance and prophylactic angioplasty improved AV graft outcomes.

A variety of methods have been validated for detection of hemodynamically significant AV graft stenosis (Table 68.3). Clinical monitoring consists of using information that is readily available from physical examination of the AV graft, abnormalities experienced during the dialysis sessions (difficult cannulation or prolonged bleeding from the needle puncture sites), or unexplained decreases in the dose of dialysis, urea reduction ratio (URR), or the volume indexed

Table 68.3 Methods of Stenosis Monitoring

Clinical Monitoring

Physical examination (abnormal bruit, absent thrill, distal edema)
Dialysis abnormalities (prolonged bleeding from needle sites, difficult cannulation)
Unexplained decrease in Kt/V or URR

Surveillance

Static dialysis venous pressure (adjusted for systemic pressure)
Access blood flow
Q_a < 600 mL/min
Q_a decreased by >25% from baseline
Doppler ultrasound

Table 68.4 Positive Predictive Value of Monitoring Methods for Graft Stenosis

Surveillance Method	N of Measurements	Positive Predictive Value
Clinical Monitoring		
Cayco, 1998 ⁸⁷	68	93%
Robbin, 1998 ⁸⁸	38	89%
Safa, 1996 ⁸⁶	106	92%
Maya, 2004 ⁸²	358	69%
Robbin, 2006 ⁸⁹	151	70%
Static Venous Pressure		
Besarab, 1995 ⁹⁰	87	92%
Flow Monitoring		
Schwab, 2001 ⁹¹	35	100%
Moist, 2003 ⁹²	53	87%
Ultrasound		
Robbin, 2006 ⁸⁹	122	80%

clearance-time product (Kt/V).^{82,83,86} The positive predictive value has been reported to be between 69% and 93% for clinical monitoring (Table 68.4).^{82,86–89} AV graft surveillance uses noninvasive tests requiring specialized equipment or technician training that are not obtained as part of the routine dialysis treatment. These include measurement of static dialysis venous pressure (normalized for the systemic pressure), measurement of the access blood flow, or Duplex ultrasound to directly evaluate stenosis. Each of these monitoring or surveillance tools has been reported to have a positive predictive value for AV graft stenosis between 70% and 100% (Table 68.4).^{89–92} The negative predictive value has not been studied systematically, as it would require obtaining routine angiography (fistulograms or “graft-o-grams”) in patients whose screening test is negative. However, the negative predictive value can be inferred from the proportion of AV graft thromboses not preceded by abnormalities of AV graft surveillance, which is around 25%.⁹³

In contrast, the predictive value of surveillance methods for AV graft *thrombosis* is much less impressive. Thus, when AV grafts with abnormal monitoring criteria suggestive of stenosis are observed without preemptive angioplasty, only about 40% of the AV grafts clot over the following three months.^{94,95}

In practice, this means that in any program of AV graft monitoring, about half of preemptive angioplasties performed may be unnecessary. Unfortunately, there are no reliable tests to distinguish between the subset of AV grafts with stenosis that will progress to thrombosis from those that will remain patent without any intervention.

Several observational studies have documented that introduction of a monitoring or surveillance program for AV graft stenosis with preemptive angioplasty lowered the frequency of AV graft thrombosis by 40% to 80%, as compared to the historical control period during which there was no monitoring program (Table 68.5).^{27,86,87,90,96,97} The promising findings from multiple observational studies led the K/DOQI Work Group to recommend implementing programs of graft surveillance and preemptive angioplasty in all dialysis centers, with the goal of reducing the frequency of AV graft thrombosis.⁶⁴

Only in the last few years has the value of AV graft stenosis surveillance been subjected to rigorous testing in randomized clinical trials. To date, there have been six such trials, evaluating surveillance with access flow monitoring, static dialysis venous pressure, or ultrasound (Table 68.6).^{89,92,98–101} Only

one of the six randomized trials has demonstrated a benefit of ultrasound AV graft surveillance¹⁰¹; the other five showed no benefit of surveillance, despite a substantial increase in the frequency of preemptive angioplasty in the surveillance group.^{89,92,98–100} For example, one study randomized patients with AV grafts to standard clinical monitoring alone or to a combination of clinical monitoring and ultrasound surveillance for stenosis. The patients in the ultrasound group underwent a 66% higher frequency of preemptive angioplasty, yet there was no difference between the two randomized groups in terms of frequency of AV graft thrombosis, time to first thrombosis, or likelihood of AV graft failure (Fig. 68.6A and 68.6B).⁸⁹ Because the randomized trials have been relatively small in size, it is possible they were inadequately powered to detect a modest benefit of AV graft surveillance. However, a recent meta-analysis of the randomized studies suggested that the benefit of surveillance with preemptive angioplasty, if present, is likely to be quite small, with a reduction of AV graft thrombosis not exceeding 23% and a reduction in AV graft failure no greater than 17%.¹⁰² A large-scale, multi-center study trial, which takes into account multiple factors involved in access thrombosis, would be required to provide a definitive answer to this controversial question. In the meantime, the value of surveillance of AV graft stenosis and preemptive angioplasty in improving graft outcomes remains controversial.^{103–106}

If underlying AV graft stenosis is an important predictor of AV graft thrombosis, why is preemptive angioplasty not

Table 68.5 Effect of Surveillance on Graft Thrombosis: Observational Studies

Reference	Surveillance Method	Thrombosis Rate (Per Graft-Years)		
		Historical Control	Surveillance Period	Percent Reduction
Schwab, 1989 ⁸⁵	Dynamic dialysis venous pressure	0.61	0.20	67
Besarab, 1995 ⁹⁰	Static dialysis venous pressure	0.50	0.28	64
Safa, 1996 ⁸⁶	Clinical monitoring	0.48	0.17	64
Allon, 1998 ²⁷	Clinical monitoring	0.70	0.28	60
Cayco, 1998 ⁸⁷	Clinical monitoring	0.49	0.29	41
McCarley, 2001 ⁹⁷	Flow monitoring	0.71	0.16	77

Table 68.6 Randomized Clinical Trials of Graft Surveillance

Reference	Surveillance Method	# Subjects		PTA/yr		Thrombosis-Free Survival at 1 yr		Cumulative Survival at 1 Year	
		con	surv	con	surv	con	surv	con	surv
Lumsden, 1997 ⁹⁸	Doppler US	32	32	0	1.5	0.51	0.47	N/A	N/A
Ram, 2003 ⁹⁹	Access flow	34	32	0.22	0.34	0.45	0.52	0.72	0.80
	Doppler US		35		0.65		0.70		0.80
Moist, 2003 ⁹²	Access flow	53	59	0.61	0.93	0.74	0.67	0.83	0.83
Dember, 2004 ¹⁰⁰	Static DVP	32	32	0.04	2.1	N/A	N/A	0.74	0.56
Malik, 2005 ¹⁰¹	Doppler US	92	97	N/A	N/A	N/A	N/A	0.73	0.93
Robbin, 2006 ⁸⁹	Doppler US	61	65	0.64	1.06	0.57	0.63	0.83	0.85

PTA, Percutaneous transluminal angioplasty; US, ultrasound; DVP, dialysis venous pressure; con, control; surv, surveillance.

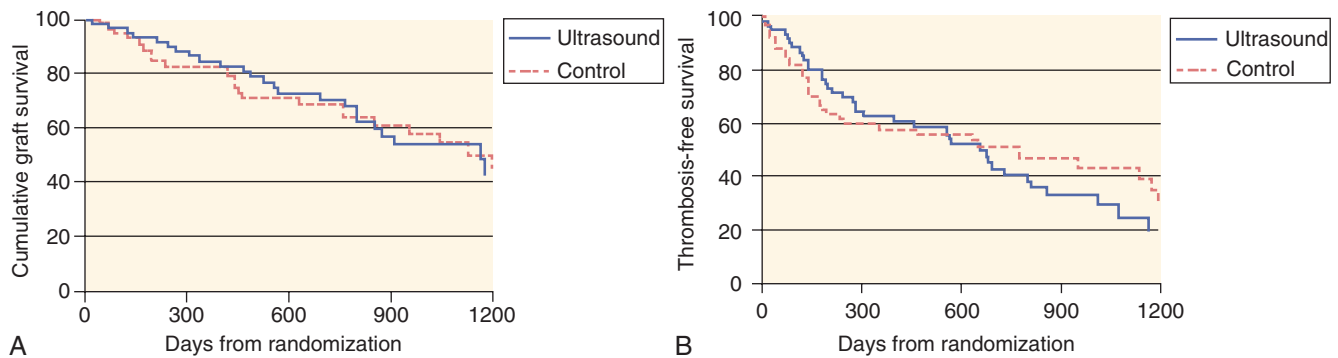


Fig. 68.6 A, Comparison of cumulative graft survival between randomized patients with clinical monitoring versus clinical monitoring plus regular ultrasound surveillance of grafts. $P = 0.93$ by the log rank test. B, Comparison of thrombosis-free graft survival between randomized patients with clinical monitoring versus clinical monitoring plus regular ultrasound surveillance of grafts. $P = 0.33$ by the log rank test. (A and B, Reproduced from Robbin ML, Oser RF, Lee JY, et al. Randomized comparison of ultrasound surveillance and clinical monitoring on arteriovenous graft outcomes. *Kidney Int.* 2006;69(4):730–735.)

more successful in reducing AV graft thrombosis? The fundamental problem appears to be the short-lived efficacy of angioplasty to relieve AV graft stenosis. When serial access blood flows have been used as a surrogate marker of successful angioplasty, 20% of AV grafts have recurrent stenosis within one week of angioplasty, and 40% within one month of angioplasty.^{91,92} In another study, the mean vascular access blood flow following angioplasty increased from 596 to 922 mL/min. However, three months later the mean flow had decreased to 672 mL/min.¹⁰⁷ In addition, there is published evidence suggesting that the injury from balloon angioplasty can actually accelerate neointimal hyperplasia, thereby resulting in recurrent stenosis.¹⁰⁸ Not surprisingly, patients undergoing angioplasty for AV graft stenosis require frequent re-interventions due to recurrent stenosis. The median intervention-free patency following AV graft stenosis is only about six months.^{82,83} Finally, AV graft surveillance is likely to be most useful in new AV grafts (within three months of their creation).¹⁰⁹ Unfortunately, a large proportion of AV grafts fail before there has been adequate opportunity for surveillance to be performed and for hemodynamically significant stenosis to be detected and treated.¹¹⁰ One study evaluated drug-coated balloon angioplasty with paclitaxel versus standard balloon angioplasty to evaluate the effects of minimizing endothelial injury following angioplasty.¹¹¹ The results from this study showed that cumulative target patency at 6 months, defined as angiographic evidence of a patent lesion or circuit with <50% angiographic restenosis and without requirement for repeat procedures during the follow-up period was significantly higher in the drug-coated balloon group (70% in drug-coated balloon group vs. 25% in standard balloon angioplasty group, $p < 0.0001$).¹¹¹ A recent multicenter, randomized controlled trial in 285 patients evaluated the safety and effectiveness of Lutonix® drug-coated balloon compared to standard balloon angioplasty in treating patients presenting with clinical and hemodynamic abnormalities in native AV fistulas located in the upper extremity.¹¹² The primary efficacy endpoint was target lesion primary patency at 6 months defined as no need for clinically driven reintervention on the target lesion or access thrombosis.¹¹² The 6-month endpoint was not met for target lesion primary patency (71%

for the drug-coated balloon and 63% for the control group; $P = 0.06$).¹¹² However, both arms of the study demonstrated equivalent safety.¹¹²

The pathophysiology of AV graft stenosis involves proliferation of vascular smooth muscle cells (neointimal hyperplasia), with progressive encroachment of the lesion into the AV graft lumen.¹¹³ To improve the patency of AV grafts following angioplasty, some investigators have attempted stent deployment. The rationale is that the rigid scaffold of the stent helps to keep open the vascular lumen.¹¹⁴ There has also been an ongoing interest in pharmacologic approaches to prevention of neointimal hyperplasia. Two small, single-center, randomized clinical trials have documented a beneficial effect of dipyridamole and fish oil in preventing AV graft thrombosis.^{115,116} A multicenter, randomized, double-blind study compared clopidogrel plus aspirin to placebo for prevention of AV graft thrombosis. The study was terminated early due to an excess of bleeding complications in the intervention group; there was no difference in the rate of AV graft thrombosis between the two randomized groups.¹¹⁷ Similarly, a randomized clinical trial found that low-intensity warfarin posed a substantial risk of major hemorrhagic complications without reducing the frequency of AV graft thrombosis.¹¹⁸

The Dialysis Access Consortium was a larger multicenter, randomized, double-blinded clinical trial comparing long-acting dipyridamole + low-dose aspirin (Aggrenox™) to placebo in prevention of AV graft failure.¹¹⁰ There was a modest but statistically significant improvement in primary unassisted AV graft survival in patients receiving dipyridamole-aspirin, as compared with those treated with placebo (28% vs 23%, at one year $P = 0.03$). Cumulative AV graft survival did not differ between the two treatment arms.¹¹⁰ Most recently, a double-blind, randomized controlled study evaluating fish oil on PTFE AV graft patency and cardiovascular outcomes reported that fish oil did not decrease loss of native patency (time from creation to first intervention) of AV grafts at 12 months, but did show improvement in overall AV graft patency and cardiovascular outcomes.¹¹⁹ More detailed discussion of pharmacologic approaches to prophylaxis of AV graft stenosis and thrombosis is beyond the scope of this chapter, but has been reviewed recently.¹²⁰

ANGIOPLASTY OF AV GRAFT STENOSIS

Patients with patent AV grafts are frequently referred for elective angioplasty when hemodynamically significant stenosis is detected either by clinical monitoring or by AV graft surveillance. The goal of elective angioplasty is to correct the stenotic lesion that impairs optimal delivery of dialysis, and hopefully delay AV graft thrombosis. The most common location of the stenosis localized by angiography is the venous anastomosis, followed by the peripheral draining vein, the central veins, and intra-graft (Table 68.7).^{82–84} Inflow (arterial anastomosis) stenosis has been rare (<5%) in most series. However, one study, using retrograde angiography with manual occlusion of the venous limb, documented an inflow stenosis (> 50%) in 29% of grafts referred for diagnostic angiography, although all these patients also had venous anastomosis stenosis as well.¹²¹

A number of published series have documented the short-lived primary patency (time to next radiologic or surgical intervention) of AV grafts following elective angioplasty (Table 68.8),^{82–84,86,122,123} with only 50% to 60% patency at 6 months and 30% to 40% at 1 year. This means that on the average, each AV graft requires two angioplasties per year. The primary patency is shorter after angioplasty of central vein stenosis, as compared with other stenotic locations. In one study, the primary patency at 6 months was only 29% for central vein stenosis, as compared with 67% for stenosis at the venous anastomosis.⁸⁴ Most studies have documented progressively shorter patency after each consecutive angioplasty, although

one investigator found comparable primary patency for the first and subsequent AV graft angioplasties.⁸⁴

The primary patency of AV grafts following elective angioplasty is not affected by patient age, race/ethnicity, diabetes, or peripheral vascular diseases.⁸² However, the patency tends to be shorter in women than in men.⁸² The primary patency after angioplasty is also not influenced by the location of the AV graft or the number of concurrent stenotic lesions found.⁸³

The technical success of an angioplasty procedure may be assessed in several ways. The first is visual inspection of the fistulogram before and after the procedure to determine whether the magnitude of the stenosis (percent stenosis relative to the normal vessel diameter) has been reduced. The degree of stenosis of each lesion can be quantified with calipers, electronic quantitative analysis (QVA) or graded semi-quantitatively using the following scale: grade 1, no (<10%) stenosis; grade 2, mild (10% to 49%) stenosis; grade 3, moderate (50% to 69%) stenosis; and grade 4, severe (70% to 99%) stenosis.^{82,83} A second approach is to measure the intra-graft pressure before and after the procedure, and normalize it for the systemic blood pressure. A third approach is to measure the change in access blood flow before and after the procedure. Each of these measures has been shown to be predictive of the primary patency of AV grafts following elective angioplasty. In one large series, following elective angioplasty, the median intervention-free survival of AV graft with no residual stenosis was 6.9 months, as compared with 4.6 months with any degree of residual stenosis.⁸³ Similarly, the median primary patency of AV grafts following angioplasty was inversely-related to the intra-graft to systemic pressure ratio, being 7.6, 6.9, and 5.6 months, when this ratio was < 0.4, 0.4 to 0.6, and >0.6, respectively.⁸³ Finally, a failure to significantly increase the access blood flow after angioplasty is observed in 20% of AV grafts at 1 week and in 40% by 1 month,^{124,125} confirming the short-lived benefit of this intervention.

Measurements of the degree of stenosis are not always feasible, nor are they entirely objective. Caliper measurement or “eyeballing” are to various degrees, subjective. Even QVA is not reliable in detecting the various densities and edges of the normal vessels and requires operator adjustment of detected vascular edges (a subjective intervention). Arterio-venous graft loops originating from the brachial artery and anastomosis at a 90-degree angle with the basilic vein are hardest to image in profile, which is necessary in order to get an accurate measurement of stenosis. We routinely obtain intra-graft pressure measurement during angiography and

Table 68.7 Location of Stenosis in Patients With Grafts Undergoing Angioplasty

Reference	Percent of All Stenotic Lesions				
	VA	VO	CV	IG	AA
Beathard, 1992 ⁸⁴	42	34	4	20	0
Lilly, 2001 ⁸³	55	22	15	6	2.5
Elective PTA					
Lilly, 2001 ⁸³	60	14	9	10	7
Thrombectomy					
Maya, 2004 ⁸²	62	16	8	12	1.5

VA, Venous anastomosis; VO, venous outflow; CV, central vein; IG, intragraft; AA, arterial anastomosis; PTA, percutaneous transluminal angioplasty.

Table 68.8 Primary Graft Patency After Elective Angioplasty

Reference	N Procedures	Primary Patency at:		
		3 Months	6 Months	12 Months
Beathard, 1992 ⁸⁴	536	79	61	38
Kanterman, 1995 ¹²²	90		63	41
Safa, 1996 ⁸⁶	90	70	47	16
Turmel-Rodrigues, 2000 ¹²³	98	85	53	29
Lilly, 2001 ⁸³	330	71	51	28
Maya, 2004 ⁸²	155	79	52	31

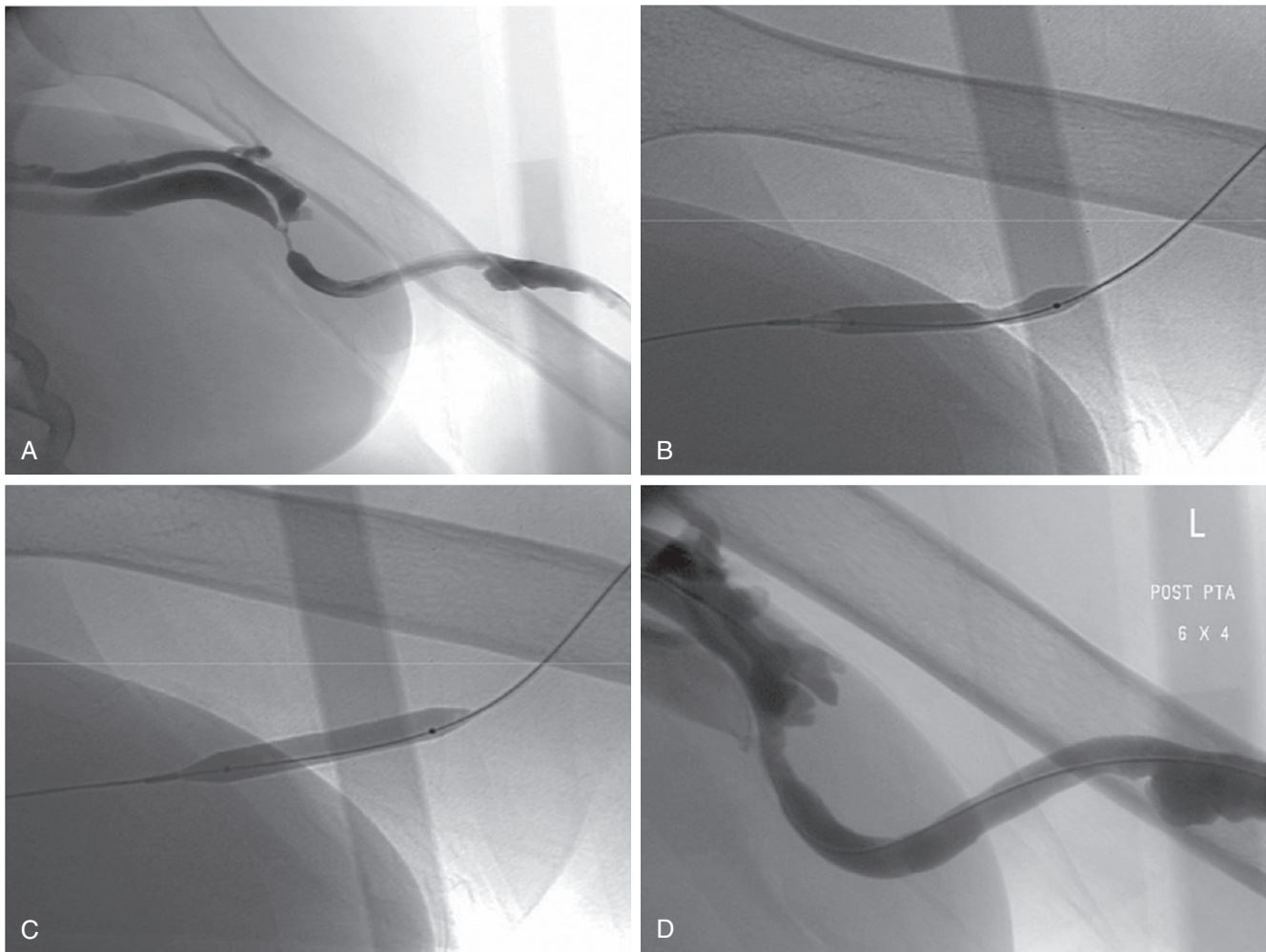


Fig. 68.7 A, Left upper arm arteriovenous (AV) graft angiogram showing a severe (95%) stenotic lesion at the level of the venous anastomosis. B, Left upper arm AV graft stenotic lesion at the venous anastomosis with the angioplasty balloon partially inflated. C, Left upper arm AV graft stenotic lesion at the venous anastomosis with the angioplasty balloon fully inflated. D, Final post-angioplasty left upper arm AV graft angiogram showing a treated lesion with minimal residual stenosis.

PTA to add a hemodynamic parameter in addition to the visual assessment of fistulograms.

TECHNICAL PROCEDURE: PERCUTANEOUS GRAFT ANGIOPLASTY

A digitally subtracted antegrade angiogram is performed to visualize the entire access circuit. The presence or absence of stenotic lesions and their number and location is assessed, and arterial anastomosis, intragraft, venous anastomosis, draining vein and central vein lesions are documented. The degree of stenosis of each lesion is quantified with calipers or graded semi-quantitatively.^{82,83} Lesions with at least 50% stenosis are considered to be hemodynamically significant and undergo angioplasty. If a stenotic lesion is encountered, then an angioplasty of the lesion or lesions is performed. Angioplasty balloon sizes vary from 4 to 14 mm in diameter to 20 to 80 mm in length, depending on the vessel to be treated. The chosen balloon is usually selected to be 1 mm larger than the size of the AV graft or the vessel to be treated. The balloon is placed within the stenotic lesion, through a sheath, and inflated to its nominal pressure for 30 to 90

seconds (Fig. 68.7A, 68.7B, 68.7C, 68.7D) and (Fig. 68.8A, 68.8B, 68.8C). The majority of anastomotic lesions require higher pressure than those required for peripheral arterial angioplasty. Therefore, high pressure balloons with minimal burst pressure > 25 atmospheres are routinely used.¹²⁶ If a residual (>30%) stenosis is found, prolonged angioplasty (5-minute inflation), higher-pressure balloons (up to 30 atmospheres), and occasionally stent or covered stent deployment may be required to treat these lesions.

The major complication of this procedure is vessel extravasation and/or rupture of the vessel during angioplasty. Deploying a covered stent (wall-graft) can treat these complications. Surgical repair is indicated if the rupture is not corrected by stent placement.

THROMBECTOMY OF ARTERIOVENOUS GRAFTS

The majority of AV graft failures are due to thrombosis, which occurs most commonly in the context of underlying stenosis at the venous anastomosis.^{83,127} For this reason, successful graft thrombectomy requires both resolution of the clot and

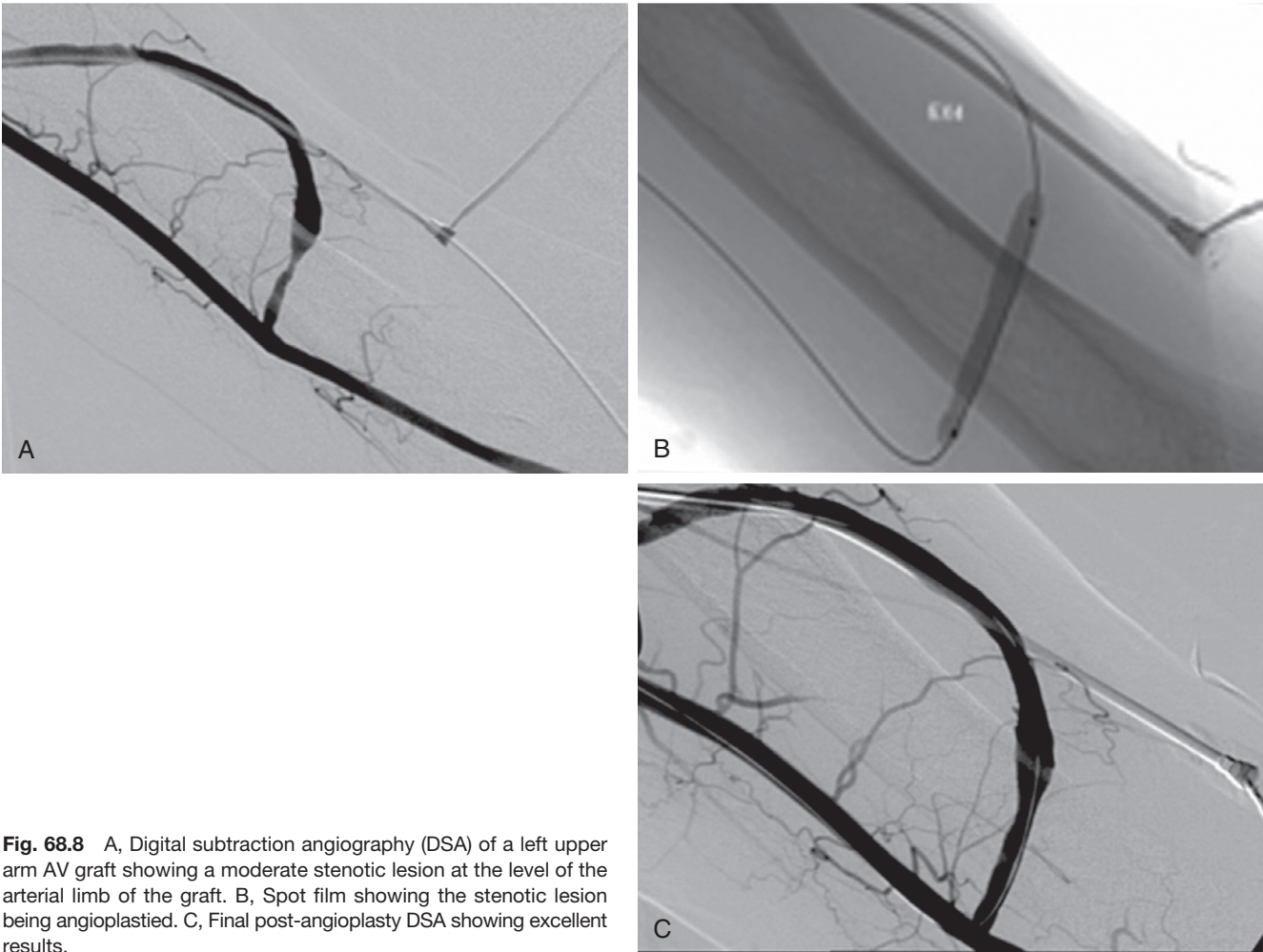


Fig. 68.8 A, Digital subtraction angiography (DSA) of a left upper arm AV graft showing a moderate stenotic lesion at the level of the arterial limb of the graft. B, Spot film showing the stenotic lesion being angioplastied. C, Final post-angioplasty DSA showing excellent results.

correction of the underlying stenotic lesion. The primary patency of AV grafts after thrombolysis or thrombectomy and angioplasty (Table 68.9) ranges from 30% to 63% at 3 months and 11% to 39% at 6 months.^{83,123,128–135} These outcomes are considerably worse than the primary patency observed after elective angioplasty (Table 68.8), which is 71% to 85% at 3 months and 47% to 63% at 6 months.^{82–84,86,122,123}

The primary AV graft patency is similar for mechanical thrombectomy and pharmaco-mechanical thrombectomy.¹³⁰ A large series comparing the outcomes of both types of radiologic procedures at one institution found that the primary patency was only 30% at 3 months for clotted AV grafts, as compared with 71% for patent AV grafts undergoing elective angioplasty (Fig. 68.9A).⁸³ The discrepancy was still apparent when the analysis was restricted to the subset of procedures in which there was no residual stenosis, with a median primary patency of 2.5 months after thrombectomy, as compared with 6.9 months after elective angioplasty (Fig. 68.9B).⁸³

The duration of AV graft patency following thrombectomy does not appear to differ among patients with and without diabetes mellitus. Arteriovenous graft patency is also generally unrelated by the AV graft location or the number of concurrent AV graft stenoses found.⁸³ However, similar to the

Table 68.9 Primary Graft Patency After Thrombectomy

Reference	N Procedures	Primary Patency at:	
		3 Months	6 Months
Valji, 1991 ¹²⁸	121	53	34
Trerotola, 1994 ¹²⁹	34	45	19
Beathard, 1994 ¹³⁰	55 mech	37	
	48 pharm	46	
Cohen, 1994 ¹³¹	135	33	25
Sands, 1994 ¹³²	71		11
Beathard, 1995 ¹³³	425	50	33
Beathard, 1996 ¹³⁴	1176	52	39
Trerotola, 1998 ¹³⁵	112	40	25
Turmel-Rodrigues, 2000 ¹²³	58	63	32
Lilly, 2001 ⁸³	326	30	19

Mech, Mechanical; *pharm*, pharmacomechanical.

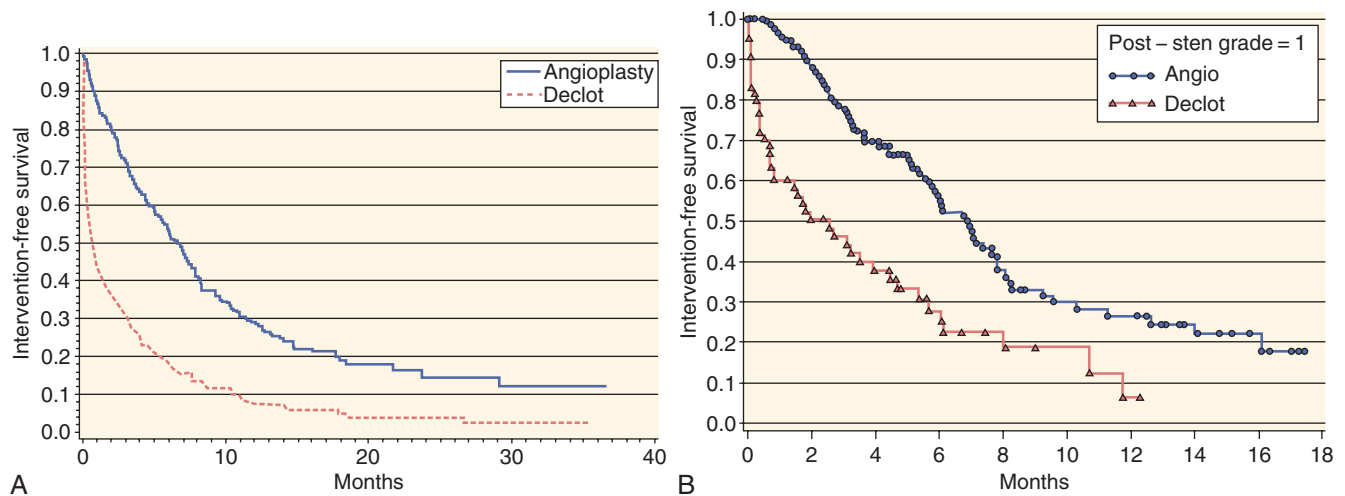


Fig. 68.9 A, Intervention-free graft survival following elective angioplasty (solid line) or thrombectomy + angioplasty (dotted line). Graft survival was calculated from the date of the initial intervention to the date of the next intervention (angioplasty, declot, or surgical revision). $P < 0.001$ for the comparison between the two groups. B, Intervention-free graft survival following elective angioplasty (circles) or thrombectomy + angioplasty (triangles) in the subset of procedures with no residual stenosis. Graft survival was calculated from the date of the initial intervention to the date of the next intervention (angioplasty, declot, or surgical revision). $P < 0.001$ for the comparison between the two groups. (A and B, Reproduced from Lilly RZ, Carlton D, Barker J, et al. Predictors of arteriovenous graft patency after radiologic intervention in hemodialysis patients. *Am J Kidney Dis.* 2001;37(5):945–953.)

observations obtained after elective angioplasty, the primary patency of AV grafts after thrombectomy is inversely proportional to the magnitude of residual stenosis at the end of the procedure.⁸³

TECHNICAL PROCEDURE: PERCUTANEOUS ARTERIOVENOUS GRAFT THROMBECTOMY

The AV graft is initially accessed with a single-wall needle at level of the arterial limb of the graft. A guidewire is passed up to the venous outlet and the needle exchanged for a 6-French sheath. A catheter is placed beyond the clotted AV graft and a venogram is performed of the venous outflow and central circulation. Extreme caution is exercised not to pressure inject contrast into the AV graft, since it can dislodge clot and cause arterial emboli. Since more than 60% of the stenotic lesions are located at the venous anastomosis, an angioplasty of this area is performed, usually with an 8-mm balloon. Mechanical thrombectomy can be achieved by one of several methods: manual aspiration of the clots, infusion of a thrombolytic agent (tissue plasminogen activator [tPA], urokinase), use of a “clot buster” device (Angiojet, Arrow-Trerotola, Cragg thrombolytic brush, Hydrolyser, Prolumen, Amplatz thrombectomy device), or a combination of any of the above (Fig. 68.10A and 68.10B). Pure mechanical thrombectomy is generally sufficient, and thrombolysis is rarely necessary. Second, entry of the AV graft is performed toward the arterial anastomosis. A Fogarty balloon is passed beyond the arterial anastomosis and pulled back to dislodge the arterial plug, if present. Aspiration of the clots through both sheaths is performed. Finally, antegrade and retrograde angiograms of the AV graft are performed to assess for patency and look for other stenotic lesions. Intra-access pressure and systemic pressure (measured inside the AV graft by occluding the venous outflow) are measured. The ratio is calculated to confirm acceptable angioplasty results. High intragraft

pressures indicate residual venous anastomotic obstruction, whereas extreme low pressures indicate obstruction at the arterial inflow.

Major complications of the thrombectomy procedure are vessel extravasations (Fig. 68.11) and rupture of the vein either because of wire manipulation or as a result of the angioplasty. Small extravasations are self-limited and could be observed; otherwise, stent deployment is the treatment of choice. Arterial emboli distal to the arterial anastomosis may occur and if encountered either intervention or surgical embolectomy is required. One interventional method to treat this complication is “back bleeding”; this refers to an occlusion of the artery before its anastomosis to the AV graft, causing retrograde blood flow that brings the clot into the AV graft. The use of a Fogarty balloon to remove the clot and the use of thrombolytic agents can also be performed to treat this complication.¹³⁶ There are very few published studies evaluating the frequency of pulmonary embolism following thrombectomy. One study has reported that pulmonary embolism is common (~35%) in patients undergoing AV graft thrombectomy.¹³⁷

DEPLOYMENT OF STENTS FOR AV GRAFT STENOSIS/THROMBOSIS

As discussed above, the primary patency of AV grafts following angioplasty is short-lived, and there is evidence that the vascular injury from angioplasty may actually accelerate neointimal hyperplasia.¹⁰⁸ In view of these considerations, there has been considerable interest in technical modifications to improve the patency of AV grafts following angioplasty. Endoluminal stents work by forming a rigid scaffold preventing elastic recoil and help to keep the vascular lumen open. Therefore, although neointimal hyperplasia recurs, a 1-mm thick layer on the wall of a stent is unlikely to cause a

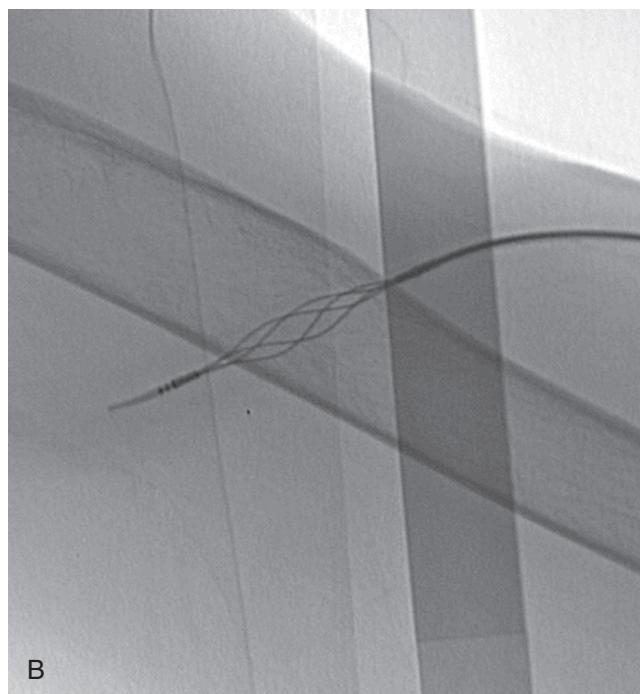
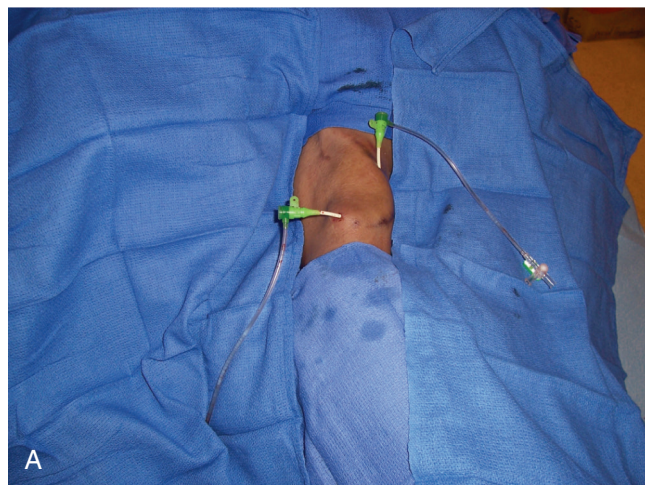


Fig. 68.10 A, Percutaneous mechanical thrombectomy: A left upper arm arteriovenous graft site is prepared and two 6-French sheaths are in place. B, Percutaneous mechanical thrombectomy: Spot film showing the use of a percutaneous thrombectomy device. (Arrow-Trerotola-PTD).

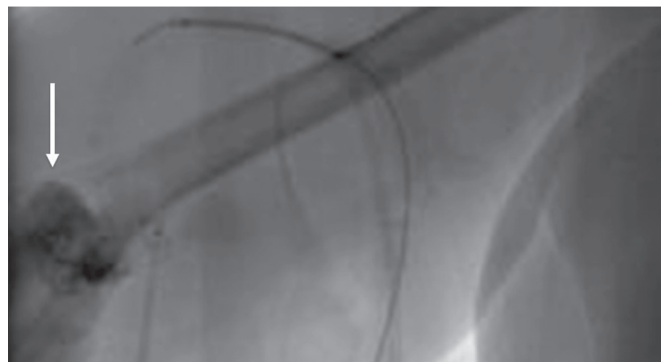


Fig. 68.11 Mechanical thrombectomy complication: Extravasation of contrast (white arrow) at the middle portion of the arteriovenous graft.

significant stenosis of the vascular lumen of an 8-mm stent. Stent placement has been attempted for the treatment of rapidly recurring stenosis. A stenosis that is highly resistant to balloon angioplasty and cannot be expanded with a balloon is a contraindication for stent placement, because the stent will be as narrow as the original stenosis. On rare occasions, when trying to overcome such resistant stenoses with very high pressure, angioplasty may result in venous rupture and extravasation. Surgery is not necessary in these situations, as the complication can be converted to success by using stents or stent grafts (endograft).

The K/DOQI Clinical Practice Guidelines for Vascular Access recommend stent deployment only when there is acute elastic recoil of the vein (>50% stenosis) after angioplasty, when stenosis recurs within a 3-month period, or to treat vascular rupture complicating the angioplasty.¹³⁸

Several small series have reported outcomes following stent deployment for refractory vascular access stenosis.^{139–145} Most of these studies have been limited by retrospective data collection, absence of a suitable control group, lumping together of patent and thrombosed accesses, and combining grafts with fistulas.¹⁴⁶ A small randomized study comparing stents with conventional angioplasty found no difference in primary AV graft patency following the intervention.¹⁴⁰ However, this study enrolled a mixture of clotted and patent AV grafts, and the stenotic lesions were at a variety of locations, thus limiting the interpretation of the findings.

Since primary AV graft patency is particularly short-lived in clotted AV grafts undergoing thrombectomy, those AV grafts may experience better patency after stent deployment. One series reported the outcomes of 34 clotted AV grafts undergoing thrombectomy with stent placement at the venous anastomosis.¹⁴³ The primary patency following intervention in this homogeneous group of AV grafts was 63% at 6 months. Although there was no matched control group treated with angioplasty alone, the primary AV graft patency was much higher than that reported previously (11% to 39% at 6 months) (Table 68.9). A nonrandomized study comparing outcomes of clotted AV grafts treated with thrombectomy and stent placement at the venous anastomosis with matched control patients treated with only thrombectomy and angioplasty observed a significantly longer primary patency in AV grafts treated with a stent, as compared with those treated with angioplasty alone.¹⁴⁷ Similarly, a retrospective study with 48 patients observed longer AV graft patency after elective treatment of stenosis with stent deployment, as compared with angioplasty alone.¹⁴⁸ A definitive, randomized clinical trial is warranted to evaluate whether the patency of AV grafts following thrombectomy is enhanced by stent deployment.

There are a number of stent types available, including covered and noncovered stents, and either balloon or self-expandable. Balloon expandable are susceptible to be crushed under pressure if used peripherally and may only be used centrally. A variety of self-expanding Nitinol-based stents are available for use outside the coronary circulation. Although they appear similar, there are subtle differences, which may favor one stent over others in a particular circulation. However, there are no published clinical trials comparing the outcomes among stent types used for dialysis access. The advent of drug-eluting stents (DES) in the coronary circulation significantly reduced recurrent stenosis from 20% to 23% to less than 10%. Drug-eluting stents were implicated in cases of acute and late thrombosis, which led to the routine and obligatory administration of antiplatelet agents (aspirin and clopidogrel) for at least one year from stenting. It is also possible that the administration of antiplatelet agents after stent placement or employment of drug-eluting stents (DES) (e.g., sirolimus) or both may further improve the primary patency of AV grafts following angioplasty and/or thrombectomy for AV grafts, but there are no definitive data at this time. However, a recent randomized clinical trial in 119 patients with failing AV fistula or AV graft randomly assigned to a paclitaxel DES or standard balloon angioplasty reported prolongation of primary patency of dialysis access stenoses at 6 months and 1 year.¹⁴⁹ Future studies should help to resolve the question of whether DES are truly superior to balloon angioplasty in managing AV graft failure.

Finally stent grafts, which consist of a metallic stent covered with graft material, have been used to treat and exclude large pseudoaneurysms in the AV graft and complications such as rupture and extravasation successfully. However, a recent multicenter, randomized clinical trial allocated 190 patients with >50% AV graft stenosis at the venous anastomosis to treatment with stent-grafts or balloon angioplasty, with follow-up protocol angiograms performed at two and six months.¹⁵⁰ AV graft patency at six months was significantly higher in patients randomized to stent-grafts, but AV graft thrombosis was not different in the two treatment arms. Another recently published study evaluated 2-year data from a multicenter randomized controlled study comparing an expanded PTFE stent graft and percutaneous transluminal angioplasty for treatment of AV graft anastomotic stenosis.¹⁵¹ Treatment area and overall access primary patency and time to subsequent intervention was longer in the stent graft group versus the angioplasty group.¹⁵¹ Despite the cost of these endografts/covered stents, they may become quite valuable if long-term patency proves superior to angioplasty and bare metal stents.

TECHNICAL PROCEDURE: PERCUTANEOUS DEPLOYMENT OF STENT

If a severe elastic recoil is seen on the final angiogram or a large residual stenosis (>50%) is seen at the level of the original stenotic lesion, a stent could be deployed.¹³⁶ Self-expanding stent sizes vary from 4 mm to 14 mm in diameter, and 10 to 80 mm in length. The most commonly used stents are nitinol or nitinol covered with expanded e-PTFE. The appropriate size and length is determined by grading the stenotic lesion at the time of placement. Usually the stent is selected to be 1 mm larger than the size of the AV graft or the vessel to be treated, and the length 5 mm longer than

the stenotic lesion on each side. The stent comes already mounted in a device that is inserted through the sheath located at the arterial limb of the AV graft. A “road map” contrast injection of the stenotic lesion is performed. The stent is placed at the site of the stenotic lesion and deployed under fluoroscopic guidance (Fig. 68.12A, 68.12B, 68.12C). A post-deployment angioplasty with an appropriate size balloon is usually necessary to reexpand recoiled lesions. A final angiogram is performed to assess for patency and proper placement of the stent.

Complications of stent deployment include those related to angioplasty. In addition, underestimation of the required stent size may result in stent migration to the systemic circulation. If the stent is placed at a site where another vessel joins the main venous outlet, that vessel may be completely or partially occluded. The stent can fracture (Fig. 68.13) and occasionally thrombose (Fig. 68.14A, 68.14B, 68.14C). Finally, a potential long-term complication is intra-stent restenosis or thrombosis, which may require multiple frequent re-interventions. There has been a recently published multicenter randomized controlled trial assessing the safety and efficacy of an expanded PTFE stent graft versus balloon angioplasty for the treatment of in-stent restenosis in the venous outflow of AV grafts and fistulas.¹⁵² Stent-graft use provided better access circuit primary patency and treatment area primary patency than balloon angioplasty when treating in-stent restenosis in patients receiving dialysis with AV grafts and fistulas.¹⁵²

PROCEDURES INVOLVING AV FISTULAS

SALVAGE OF IMMATURE AV FISTULAS

As compared with AV grafts, AV fistulas require a much lower frequency of intervention (angioplasty or thrombectomy) to maintain their long-term patency for dialysis.⁷⁹ However, AV fistulas have a substantially higher primary failure rate (AV fistulas that are never usable for dialysis). Similar to AV grafts, the rate of angioplasty use from USRDS data has increased from 0.16 to 0.47 (rate per patient year) from 1998 to 2007,¹ likely due to the increasing frequency of treating primary AV fistula failures. The proportion of AV fistulas with primary failure has ranged from 20% to 50% in multiple recent series, even when routine preoperative vascular mapping has been employed.³⁵ A recent landmark multicenter randomized controlled trial reported a primary AV fistula failure rate of 60% in the United States.³⁹ Primary AV fistula failures fall into two major categories: early thrombosis and failure to mature.^{153–155} Early thrombosis refers to AV fistulas that thrombose within three months of their creation, before they have been used for dialysis. Failure to mature refers to AV fistulas that never develop adequately to be cannulated reproducibly for dialysis. Non-maturation is less common with upper arm than forearm AV fistulas.³⁵ Among upper arm AV fistulas, non-maturation is less likely with transposed brachiocephalic AV fistulas than with brachiocephalic AV fistulas.¹⁵⁶

In some patients, maturation of the AV fistula can be assessed easily by clinical evaluation by the nephrologist, surgeon, or an experienced dialysis nurse. In less straightforward cases, Duplex ultrasound may be useful in predicting

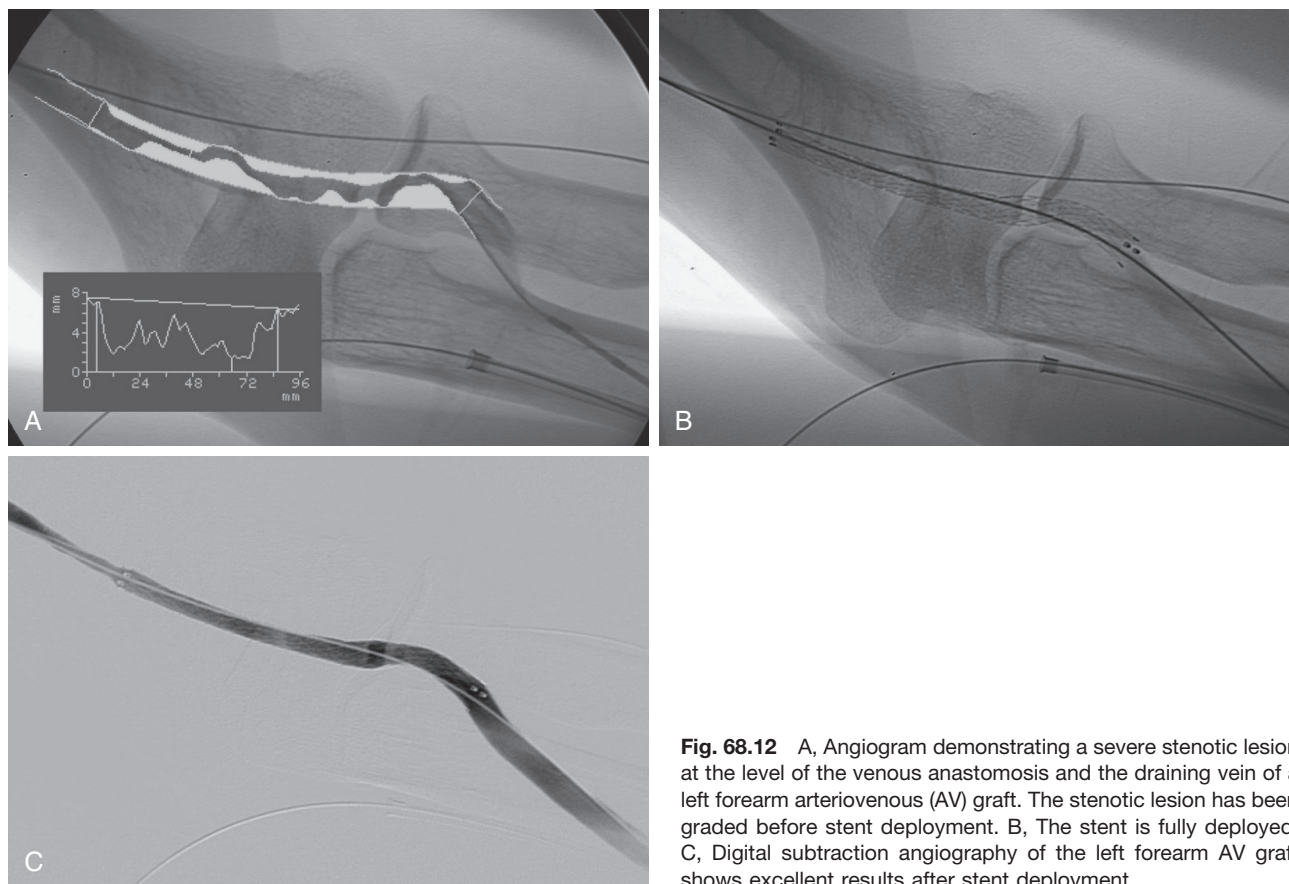


Fig. 68.12 A, Angiogram demonstrating a severe stenotic lesion at the level of the venous anastomosis and the draining vein of a left forearm arteriovenous (AV) graft. The stenotic lesion has been graded before stent deployment. B, The stent is fully deployed. C, Digital subtraction angiography of the left forearm AV graft shows excellent results after stent deployment.

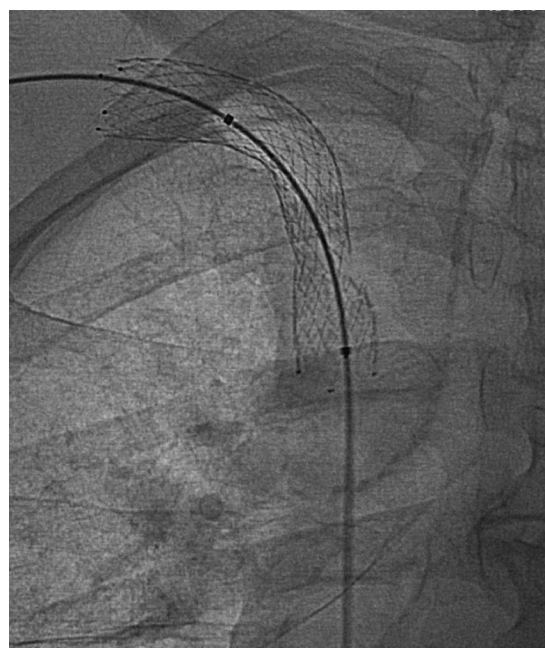


Fig. 68.13 Stent fracture at the level of the right subclavian vein and right innominate vein.

whether a new AV fistula can be used successfully for hemodialysis. One pilot study used a combination of two simple sonographic criteria to assess AV fistula maturation: AV fistula diameter and access blood flow.⁴ When the ultrasound documented a draining vein diameter ≥ 4 mm and an access blood flow ≥ 500 mL/min, 95% of the AV fistulas were subsequently usable for dialysis. In contrast, when neither criterion was met, only 33% of the AV fistulas achieved adequacy for dialysis. The likelihood of AV fistula adequacy for dialysis was intermediate (~70%) when only one of the two criteria was met.

Failure of an AV fistula to mature can be related to one of several anatomic defects, which can be identified either by sonography or angiography.¹⁵⁷ Stenosis at the anastomosis or in the draining vein is one such problem. Another possibility is that the draining vein has one or more large side branches. With these tributary veins, the arterial blood flow is distributed among two or more competing veins, thereby limiting the increase in blood flow in each. A third scenario may be observed in obese patients, in which the AV fistula has adequate caliber and blood flow, but is simply too deep to be cannulated safely by the dialysis staff. In most patients, these anatomic problems can be corrected by percutaneous or surgical interventions. Stenotic lesions can be treated by angioplasty or surgical revision. Superficial side branches can be ligated by a suture through the skin; deeper branches can be embolized. Finally, the surgeon can superficialize AV fistulas that are too deep to be cannulated safely.

In immature AV fistulas with one or more of these anatomic lesions, specific interventions to correct the underlying lesion

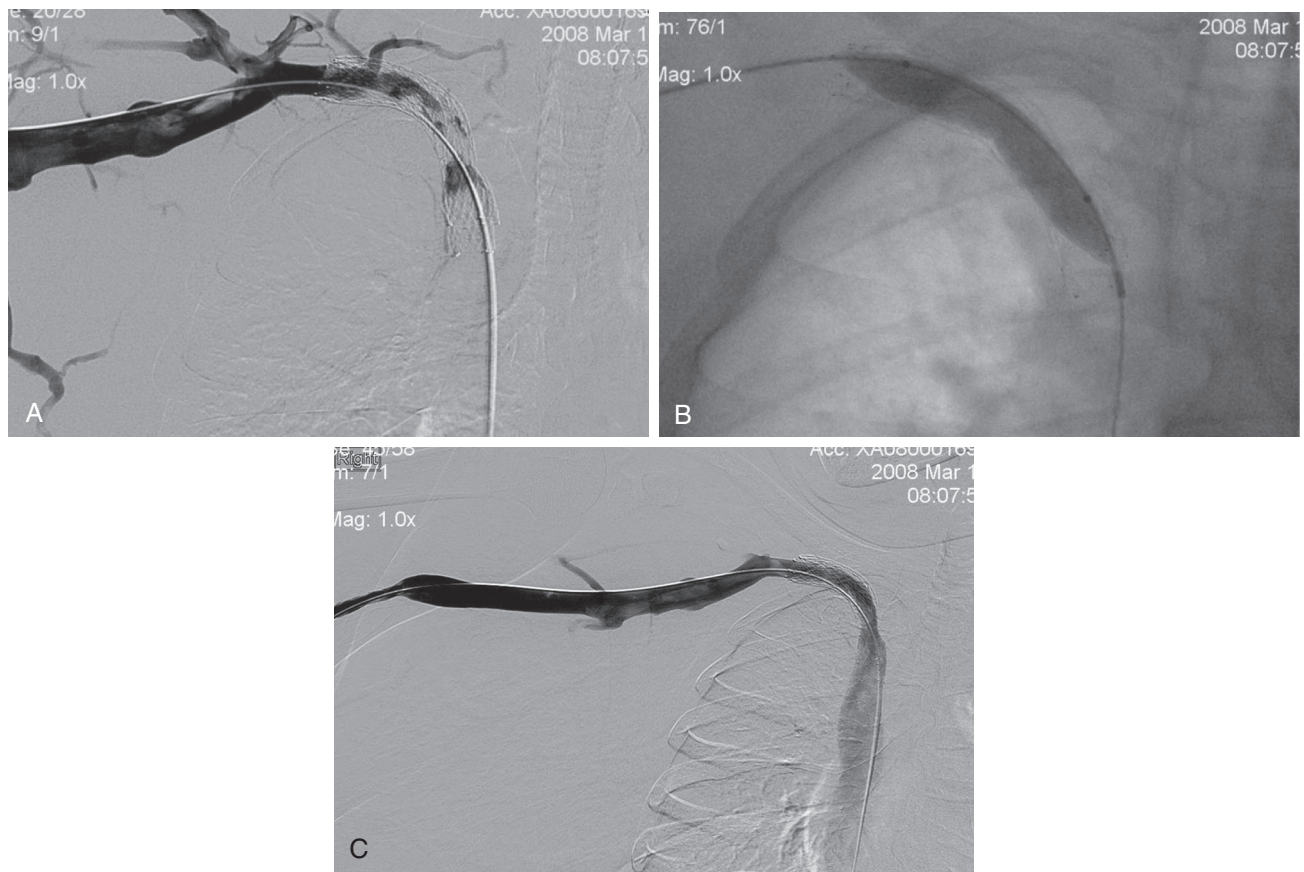


Fig. 68.14 A, Thrombosed and fractured stent at the level of the right subclavian vein and right innominate vein. B, Angioplasty of the stenotic intra-stent lesion, which caused the thrombosis of the access circuit. C, Restoration of blood flow of the thrombosed stent.

may promote subsequent AV fistula maturation. Several published series have evaluated the ability to salvage immature AV fistulas such that they are subsequently usable for dialysis. A number of studies utilizing only radiographic procedures (angioplasty of stenotic lesions or obliteration of side branches) in immature AV fistulas have had a high success rate (Table 68.10).^{77,158–162} An initial salvage (ability to use the AV fistula for dialysis) was accomplished in 80% to 90% of patients, with a subsequent 1-year primary patency of 39% to 75%. In another study utilizing a combination of radiologic and surgical salvage procedures in an unselected dialysis population, the salvage rate was more modest at 44%.¹⁵⁵ Of interest, the frequency of a salvage procedure for immature AV fistulas in that study was twice as high in women than in men. A recent retrospective study reported that immature AV fistulas with an anatomic abnormality that was corrected percutaneously or surgically were much more likely to achieve suitability for dialysis, as compared with similar fistulas that did not undergo a salvage procedure.¹⁶³ Recently, a novel approach to enhance AV fistula maturation, balloon assisted maturation (BAM), has been described.¹⁶⁴ In this procedure, repeated long-segment angioplasty procedures are used to sequentially dilate the peri-anastomotic venous segment, converting it at times into a collagen tube.¹⁶⁴ Other recent articles have described the use of intraoperative “primary balloon angioplasty” at the time of surgery $\pm$ BAM, which could allow for AV fistulas to be created in patients with

small arteries and veins (artery, <2 mm; vein, <2.5 mm).^{165,166}

None of these studies have evaluated their techniques with a control group, and all have suggested that a large number of post-maturation angioplasties are needed to maintain patency. There are no randomized studies comparing different types of interventions or different timing of interventions in terms of their effect on AV fistula maturation. Two studies observed that immature AV fistulas requiring interventions to achieve maturity had significantly shorter cumulative patency and required more interventions to maintain patency, as compared with AV fistulas achieving maturation without interventions.^{167,168} (Fig. 68.15). In summary, proactive percutaneous or surgical interventions are worthwhile, as they can often convert immature AV fistulas to ones that are suitable for dialysis.

There has also been interest in pharmacologic interventions to promote AV fistula maturation. A randomized study of 877 US patients undergoing creation of a new AV fistula allocated patients to receive clopidogrel or placebo for the initial 6 weeks.³⁹ Clopidogrel significantly decreased the frequency of AV fistula thrombosis within 6 weeks. However, there was no improvement in the likelihood of clinical AV fistula maturation.³⁹ Subsequently, another randomized study from Australia and New Zealand of 567 patients reported that neither fish oil nor low-dose aspirin prevented new AVF failure.¹⁶⁹ There are several ongoing clinical trials evaluating novel therapies to improve AV fistula maturation. Perivascular elastase, topically

Table 68.10 Effect of Salvage Procedures on Immature Fistulas

Reference	# Patients	Type of Intervention	Percent Usable for Dialysis	Primary Patency at 1 Year
Beathard, 1999 ¹⁵⁸	63	PTA, Vein ligation	82	75%
Turmel-Rodrigues, 2001 ¹⁶⁰	69	PTA, Vein ligation	97	39%
Miller, 2003 ¹⁵⁵	41	PTA Vein ligation Surgical revision	44	N/A
Beathard, 2003 ¹⁵⁹	100	PTA, Vein ligation	92	68%
Asif, 2005 ⁷⁷	24	PTA, Vein ligation	92	N/A
Nassar, 2006 ¹⁶¹	119	PTA, Vein ligation	83	65%
Singh, 2008 ¹⁶³	32	PTA Vein ligation Surgical revision	78	NA
Han, 2013 ¹⁵²	141	PTA	87	72%

NA, Not available; PTA, percutaneous transluminal angioplasty.

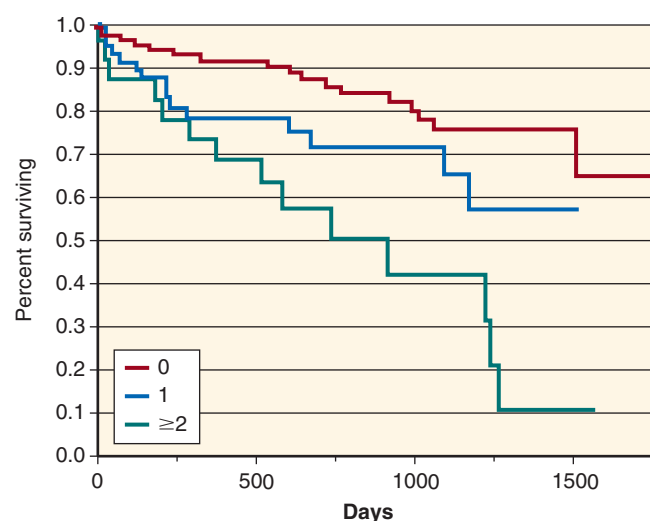


Fig. 68.15 Interventions to Promote AV Fistula Maturation: Cumulative survival, defined from the time of access cannulation to permanent failure, is shorter in patients receiving two or more interventions before arteriovenous fistula maturation compared with those with zero interventions (HR, 2.07; 95% CI, 1.21 to 2.94; $P = 0.0001$). (Reproduced from Lee T, Ullah A, Allon M, et al: Decreased Cumulative Access Survival in Arteriovenous Fistulas Requiring Interventions to Promote Maturation. *Clin J Am Soc Nephrol*. 6:575–581, 2011 with permission from the American Society of Nephrology.)

applied over the AVF artery and vein anastomosis at the time of AV fistula, has been studied in early-phase clinical trials.^{170,171} These early-phase studies of elastase to date have demonstrated safety and feasibility and have shown promising results with regard to improvement of physiologic maturation from ultrasound studies in radiocephalic AV fistulas. Given the promising results from early-phase studies, an ongoing

phase III randomized controlled clinical trial is evaluating perivascularly delivered elastase administered at the time of new radiocephalic AVF creation (NCT02110901). Finally, an ongoing randomized controlled clinical trial is evaluating locally applied nitroglycerin ointment over the area of future AV fistula creation as a therapy to enhance local nitric oxide bioavailability and improve AV fistula maturation (NCT02164318). The goal of this therapy is to restore local endothelial function in patients with CKD and those with ESKD planned for AV fistula creation in order to promote outward AV fistula remodeling and maturation.

TECHNICAL PROCEDURE: SALVAGE OF IMMATURE AV FISTULAS

a. Angioplasty of Stenotic Lesions

The stenosis at the juxta arterial anastomosis can be treated with sequential balloon dilatations. This requires two to five treatments until the size of the anastomosis is appropriate. Long segments of stenotic lesions at the level of the most proximal part of the venous outlet near the anastomosis are amenable to balloon angioplasty and sometimes may require several follow-up interventions.

Immature AV fistulas are by definition small and could be difficult to access; therefore, it is best to evaluate the vessels with high-resolution ultrasound guidance to avoid repeated puncture, extravasation, and vein damage. Subtracted digital angiograms of the venous outlet, central circulation, and a reflux retrograde arteriogram are performed. Once the lesion is identified the proper technique is selected. The initial fistulogram could be avoided if reliable information is obtained from a surveillance ultrasound indicating the presence of a stenosis near or at the arterial anastomosis.

Depending on the severity of the stenosis a balloon is selected from sizes between 2 mm through 6 mm in diameter and from 10 mm through 40 mm in length. The angioplasty

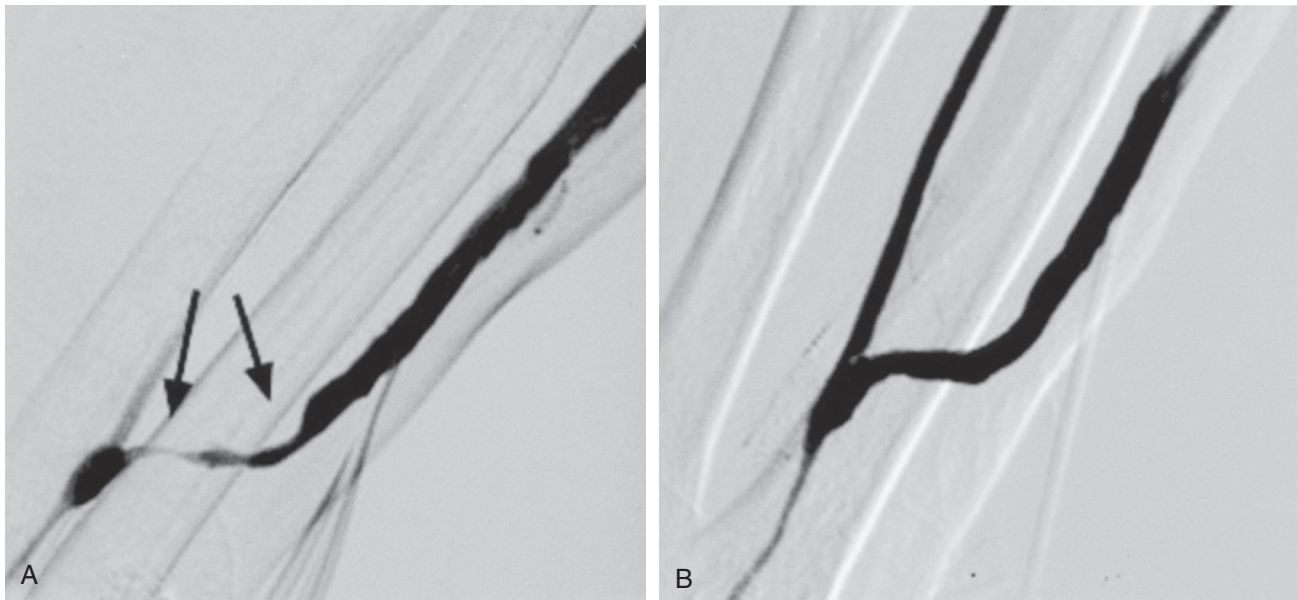


Fig. 68.16 A, Fistula salvage: Digital subtraction angiography (DSA) of a radiocephalic arteriovenous fistula showing a severe stenotic lesion at the level of the juxta arterial anastomosis. B, Fistula salvage: Post-angioplasty DSA of a radiocephalic fistula showing radiologic improvement of the stenotic lesion at the level of the juxta arterial anastomosis.

balloon is introduced into the stenotic lesion and inflated carefully up to its nominal pressure. Subsequent angiograms are performed for post-angioplasty grading of the lesion (Fig. 68.16A, 68.16B). Patients are brought back to the intervention suite two to four weeks later for a second-look angiogram of the AV fistula.

Patients are followed at their local dialysis center and if the AV fistula does not mature or problems cannulating the AV fistula persist, a follow-up angiogram of the AV fistula is indicated.

b. Ligation of Accessory Veins

Accessory veins can be treated either by surgical ligation or by endovascular coil deployment. Treatment of these lesions requires a well-trained interventionist due to the difficult technical approach to these lesions. Accessory veins are treated depending on size, location, and number (Fig. 68.17). Some interventionists advocate percutaneous ligation of superficial accessory veins at time of the initial angiogram of the AV fistula.¹⁷² If the accessory vein is deep and has a large lumen size, then surgical ligation is indicated. If the accessory vein is deep but has a small lumen size then a coil deployment should be considered (Fig. 68.18).

The AV fistula is accessed and depending on its size, an appropriate sheath is introduced. A selective catheter is introduced in each accessory vein, and an appropriate size coil is deployed. A final angiogram of the AV fistula is taken to ascertain proper coil deployment and occlusion of all collateral veins.

PERCUTANEOUS TRANSLUMINAL ANGIOPLASTY (PTA) OF AV FISTULAS

Although the frequency of interventions is several-fold lower in AV fistulas than grafts,³⁵ fistulas are also susceptible to developing stenosis and thrombosis. Most studies have

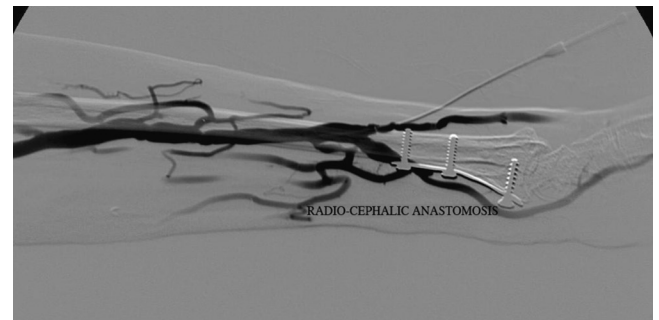


Fig. 68.17 Digital subtraction angiography of a left radiocephalic arteriovenous fistula showing multiple collaterals. There is a metallic plate from a prior open reduction and internal fixation of a radius bone fracture.

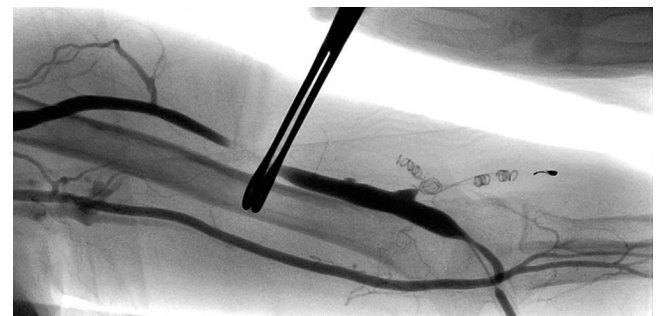


Fig. 68.18 Coil deployed in a collateral vein of an upper arm arteriovenous fistula.

documented a comparable primary patency of fistulas and grafts following elective PTA (Table 68.11),^{82,86,97,123} although one study observed a higher primary patency in fistulas.¹⁷³ As is the case with angioplasty of grafts, the primary patency of AV fistulas following PTA is inversely related to the

magnitude of post-angioplasty stenosis, as well as the magnitude of the post-angioplasty intra-access to systemic pressure ratio.⁸² It should be noted that highly matured and developed AV fistulas dilate and elongate significantly over the years, forming aneurysmal dilatation. With fixed points at the wrist and elbow or elbow and shoulder, they become tortuous and form significant kinks at certain points between dilated segments at certain intervals, which appear as significant stenoses. Many of these are not flowlimiting, while some are truly flow limiting but are unlikely to respond to PTA and may require surgical intervention. Clinical factors and major comorbidities such as age, race, diabetes, presence of peripheral vascular disease, access location, and number of stenotic sites have not been associated with the likelihood of vascular access patency after PTA.⁸²

TECHNICAL PROCEDURE: ANGIOPLASTY OF AV FISTULAS

Initial digitally subtracted angiogram of the AV fistula is performed including the venous outlet and central circulation. Lesions with more than 50% stenosis are considered hemodynamically significant and should undergo angioplasty. A

final digitally subtracted angiogram is performed to assess for residual stenosis and requirement for further treatment of the stenotic lesion (Fig. 68.19A, 68.19B, 68.19C). The patient's intra-fistula pressure and systemic pressure are measured before and immediately after the intervention, and a reduction intra-fistula to systemic pressure ratio is used to confirm hemodynamic improvement.

The major complications of this procedure are vessel extravasations and rupture of the vessel after the angioplasty (Fig. 68.20A, 68.20B). Deploying a covered stent can treat these complications. Surgical repair is indicated if the rupture is not corrected by stent placement.

PERCUTANEOUS MECHANICAL THROMBECTOMY AND THROMBOLYSIS OF AV FISTULAS

Salvaging thrombosed AV fistulas is one of the most challenging aspects in interventional nephrology.¹⁷⁴ Thrombectomy of aneurysmally dilated AV fistulas is the most difficult technically. The most common cause of AV fistula thrombosis is an underlying stenotic lesion in the venous outflow circulation (either peripherally or centrally). Less common causes include needle infiltration,¹⁷⁵ excessive manual pressure for hemostasis at the needle insertion site, or severe and prolonged hypotension.¹⁷⁶ Successful restoration of patency in a thrombosed AV fistula requires expeditious thrombectomy within 48 hours of thrombosis. Several series have reported on the outcomes of radiologic thrombectomy of AV fistulas.^{123,177–184} The immediate technical success has been fairly high, ranging from 73% to 93%. The primary patency of these AV fistulas following thrombectomy has ranged from 27% to 81% at 6 months, and 18% to 70% at 1 year (Table 68.12).^{123,178,181–185} In one study primary patency following thrombectomy was lower in upper arm AV fistulas, as compared to those in the forearm.¹²³ However, with additional interventions, the secondary patency of these AV fistulas ranges from 44% to 93% at 1 year. A recent study evaluated

Table 68.11 Primary Patency After Elective Angioplasty: Fistulas vs Grafts

Reference	Primary Access Patency at 6 Months	
	Grafts	Fistulas
Safa, 1996 ⁸⁶	43%	47%
Turmel-Rodrigues, 2000 ¹²³	53%	67%
McCarley, 2001 ⁹⁷	37%	34%
Van der Linden, 2002 ¹⁷³	25%	50%
Maya, 2004 ⁸²	52%	55%

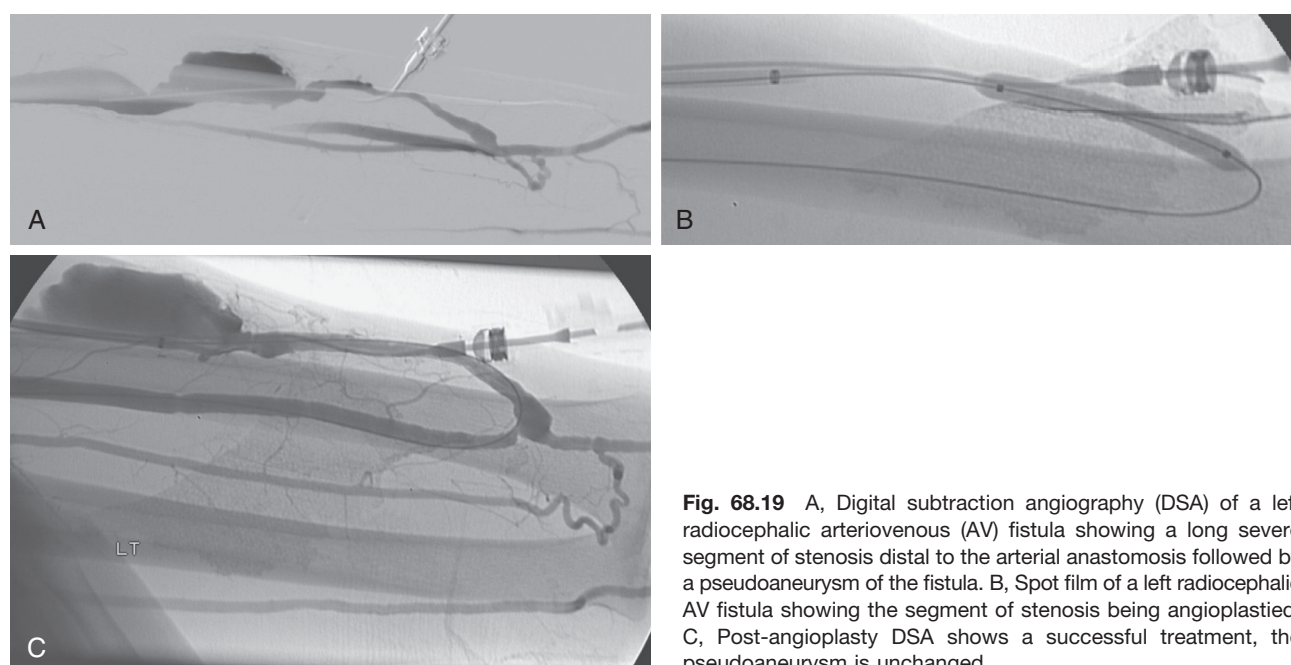


Fig. 68.19 A, Digital subtraction angiography (DSA) of a left radiocephalic arteriovenous (AV) fistula showing a long severe segment of stenosis distal to the arterial anastomosis followed by a pseudoaneurysm of the fistula. B, Spot film of a left radiocephalic AV fistula showing the segment of stenosis being angioplastied. C, Post-angioplasty DSA shows a successful treatment, the pseudoaneurysm is unchanged.

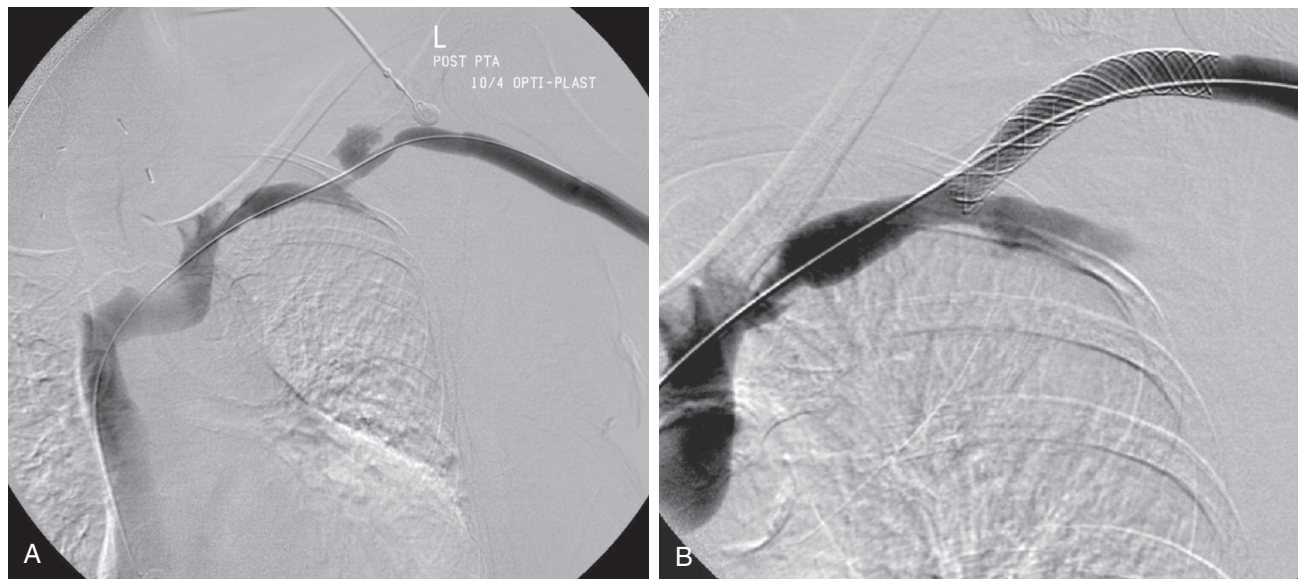


Fig. 68.20 A, Angioplasty complication: Digital subtraction angiography (DSA) showing a rupture of the left cephalic vein after aggressive percutaneous transluminal angioplasty. There is a coexisting stenosis of the left subclavian. B, Angioplasty complication: DSA showing salvage and correction of the complication by deploying a covered stent (wall-graft).

Table 68.12 Primary Fistula Patency After Thrombectomy

Reference	# Procedures	Primary Patency at:	
		6 Months	12 Months
Haage, 2000 ¹⁷⁸	54		27
Turmel-Rodrigues, 2000 ¹²³	54 FA 9 UA	74 27	47 27
Rajan, 2002 ¹⁸²	30	28	24
Liang, 2002 ¹⁸¹	42	81	70
Shatsky, 2005 ¹⁸³	44	38	18
Jain, 2008 ¹⁸⁴	41	20	—
Miller, 2011 ¹⁸⁵	140	—	59

Series with fewer than 25 procedures not included.
FA, Forearm; UA, upper arm.

140 consecutive patients, over a two-year period, with thrombosed immature AV fistulas, who underwent attempts at salvage via thrombectomy procedures.¹⁸⁵ All AV fistulas had thrombosed following access creation and had never been used for hemodialysis. Thrombectomy was successful in 119 (85%) immature clotted AV fistulas, and hemodialysis adequacy was achieved in 111 (79%) AV fistulas. The average maturation time from thrombectomy to cannulation for dialysis was 46.4 days, with an average of 2.64 interventions per patient. There were five (3.5%) cases of angioplasty-induced rupture, all of which were treated with stent placement. Clinically significant pseudoaneurysm formation occurred in four (2.8%) patients. At 12 months, secondary access patency of salvaged accesses was 90%. Considering that the alternative would be to abandon the thrombosed fistula and proceed with placement of a new AV fistula, concerted efforts to salvage thrombosed AV fistulas are extremely worthwhile.

TECHNICAL PROCEDURE: PERCUTANEOUS THROMBECTOMY OF AV FISTULAS

Although more challenging than AV graft thrombectomy, AV fistulas can also be declotted successfully. There are few contraindications, including concurrent infection, fistula immaturity, and very large aneurysms. The technical challenges include difficulty in the initial cannulation of a thrombosed AV fistula, complete removal of large thrombi, and successful treatment of recalcitrant stenotic lesions.

A percutaneous thrombectomy, an interventional procedure of removing a blood clot (thrombus) using an over-the-wire device can be used (such as Angiojet). After successful thrombolysis (dissolving of the thrombus) of the AV fistula, a Fogarty balloon is passed beyond the arterial anastomosis and pulled back to dislodge the plugging clot. Once the thrombus is cleared and blood flow reestablished, a digitally subtracted angiogram of the AV fistula is performed to evaluate for stenotic lesions along the venous outlet track or central circulation.

The major complications of this procedure are vessel extravasations and rupture of the vessel after the angioplasty. Pulmonary embolism is generally of greater concern with AV fistula thrombectomy, as compared with AV graft thrombectomy, due to the larger volume of thrombus. Finally, arterial emboli distal to the arteriovenous anastomosis may occur with higher frequency than for AV grafts.

NOVEL TECHNIQUES FOR TREATMENT OF SEVERE STENOTIC LESIONS

CUTTING BALLOON

Despite the use of angioplasty with high-pressure balloons and prolonged inflations, some lesions remain severely stenotic. The use of cutting balloons has been advocated as a tool to treat these lesions by creating a controlled rupture of the vessel wall. The cutting balloon catheter is a balloon

with four blades arranged along the balloon. When the balloon is inflated, it exposes the blades to the offending lesion; this will create a controlled rupture of the intima or hyperplastic fibrous tissue. A regular angioplasty balloon can be used afterwards to shape the vessel and expand it to the desired diameter. It has been used in lesions at all levels from intra-graft to central lesions. Preliminary reports suggested that cutting balloons may result in superior outcomes as compared with conventional angioplasty.^{186,187} In one series of nine patients, grafts with high-grade venous anastomosis stenosis were treated with cutting balloon plus stent deployment.¹⁸⁶ The patients maintained a functional graft during short-term follow-up (up to 20 months in one patient). However, a randomized, multicenter clinical trial comparing use of the cutting balloon with conventional angioplasty for treatment of AV graft stenosis observed no advantage to the cutting balloon.¹⁸⁸ The primary patency at 6 months was 48% in grafts treated with a cutting balloon, as compared with 40% for AV grafts treated with angioplasty.¹⁸⁸ Device-related complications occurred in 5% of the patients in the cutting balloon group (primarily vein rupture or dissection), as compared with none of the patients whose AV grafts were treated with angioplasty alone.¹⁸⁸ The considerable additional cost of cutting balloons is substantial, and precludes their routine use.

CRYOPLASTY BALLOON

Cryotherapy with the cryo-balloon is a novel therapy for patients with intractable stenoses at the venous anastomosis of AV grafts. This technique utilizes cold temperatures at the balloon site to cause apoptosis of the intima layer. Rifkin et al. reported the outcomes of five patients with recurrent stenotic lesions at the venous anastomosis that were treated with the cryo-balloon.¹⁸⁹ The primary patency increased from 3 weeks after angioplasty alone to more than 16 weeks after cryoplasty. There are no published randomized trials comparing the outcomes of AV graft stenosis treated with cryoplasty compared with angioplasty alone.

LOCAL PERIVASCULAR MODALITIES

Recent investigations have evaluated prophylaxis of access stenosis in animal models using local perivascular drug delivery to the site of neointimal hyperplasia. Antiproliferative drugs are injected locally during access creation, or introduced by drug-eluting stents or wraps. These novel pharmacologic therapies remain investigational, but offer promise for the future. Detailed discussion of these potential therapies is beyond the scope of this chapter, but this topic has been recently reviewed.^{120,190}

ENDOVASCULAR PROXIMAL FOREARM ARTERIOVENOUS FISTULA FOR HEMODIALYSIS ACCESS

Recently creation of an AV fistula using an endovascular approach has been evaluated in pilot studies.¹⁹¹ The rationale is that creating an AV fistula using an endovascular approach may reduce vessel trauma and injury, leading to decreased intimal hyperplasia and AV fistula maturation failure. The endovascular approach for AV fistula creation uses a novel magnet-based endovascular technology (endovascular AV fistula [endoAVF]). The technical aspects have been detailed

in two publications by Rajan et al.¹⁹² and Lok et al.¹⁹³ Early results from a prospective, single-arm, multicenter study (Novel Endovascular Access Trial (NEAT)) in 80 patients have demonstrated: (1) successful endovascular AV fistula creation in 98% of participants, (2) 87% physiologic suitability for dialysis (e.g., mean brachial artery flow, 918 mL/min; endoAVF vein diameter, 5.2 mm of cephalic vein), (3) EndoAVF functional usability of 64% in participants who received dialysis, and (4) 12-month primary and cumulative patency of 69% and 84%, respectively.¹⁹²

CENTRAL VEIN STENOSIS

Central vein stenosis is a frequent occurrence in patients on hemodialysis.^{6,194,195} Acute or chronic trauma of the central vessels caused by either temporary or permanent dialysis catheters is the major cause.¹⁹⁶ Stenosis leads to impairment of venous return on the ipsilateral extremity, and may in turn result in malfunction or thrombosis of the vascular access.¹⁹⁷ Although it may be asymptomatic, patients with central vein stenosis most commonly present with ipsilateral upper extremity edema. In some patients, a previously unappreciated central vein stenosis becomes evident clinically following creation of an ipsilateral AV fistula or graft. The diagnosis can be confirmed by angiography, ultrasound, or magnetic resonance venography.

The most commonly encountered location of central vein stenosis is at the junction of the cephalic vein with the subclavian vein (not catheter injury related). Other central veins that may be affected (often related to injury from previous catheters) include the subclavian vein, brachiocephalic vein, and superior vena cava (Fig. 68.21). In patients with tunneled femoral catheters, central vein



Fig. 68.21 Digital subtraction angiography showing a severe stenosis of the left innominate vein. There are multiple ipsilateral and across the neck collaterals draining into a normal right innominate vein.

stenosis may occur in the external iliac vein, common iliac vein or inferior vena cava (Fig. 68.22), resulting in ipsilateral lower extremity edema. The stenotic lesion is an aggressive neointimal proliferation, or clot and fibrin sheath formed around an indwelling dialysis catheter that becomes organized and incorporated into the vessel wall. Stenoses may

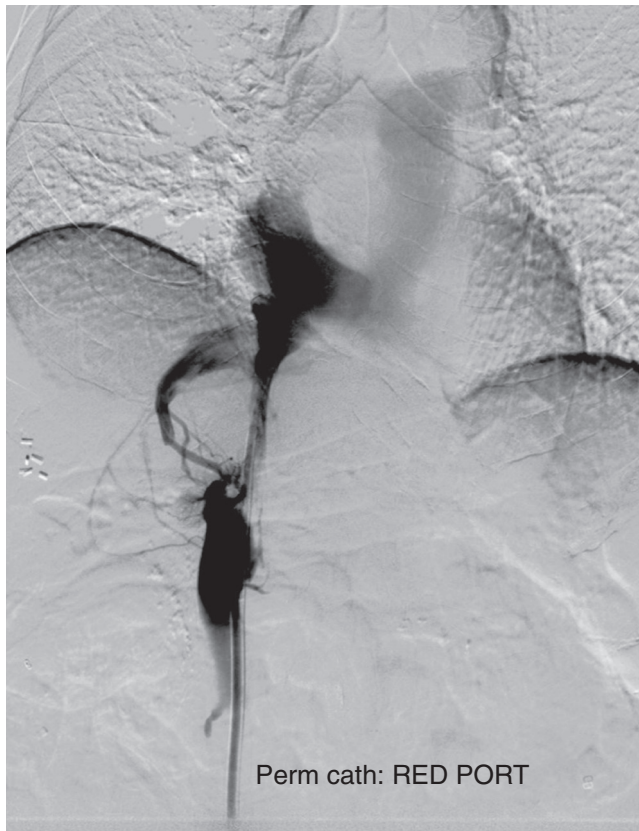


Fig. 68.22 Digital subtraction angiography showing a severe stenosis of the intrahepatic portion of the inferior vena cava with some intrahepatic collaterals.

progress over time to complete occlusion of the venous circulation (Fig. 68.23). If left untreated, central vein stenosis will cause increased retrograde pressure and formation of venous collaterals (Fig. 68.24A, 68.24B). In some patients, the collaterals are sufficiently well developed to permit adequate venous drainage that prevents formation of edema (Fig. 68.25A, 68.25B).

The treatment of choice of symptomatic central vein stenosis is PTA of the stenotic lesion.^{198–205} Unfortunately, the long-term success of PTA of central venous stenosis is quite poor, due to a combination of elastic recoil and aggressive neointimal hyperplasia. In one study, the primary patency was substantially shorter after angioplasty of central vein

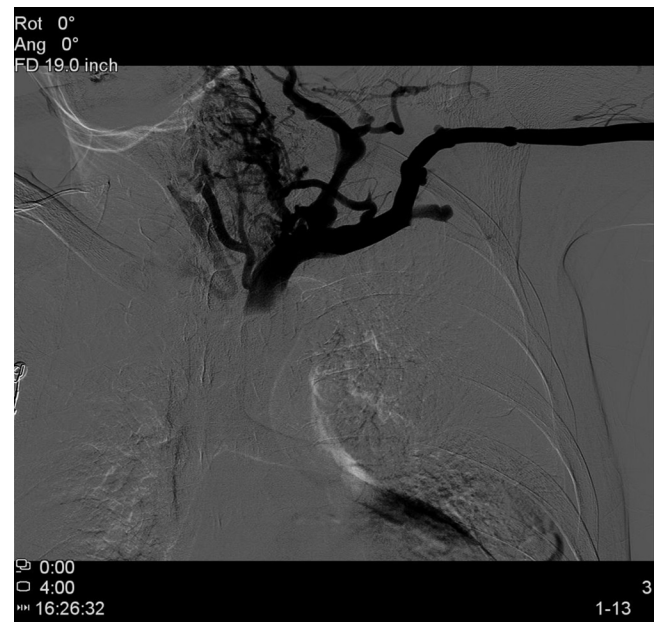


Fig. 68.23 Digital subtraction angiography showing a complete occlusion of the left innominate vein.

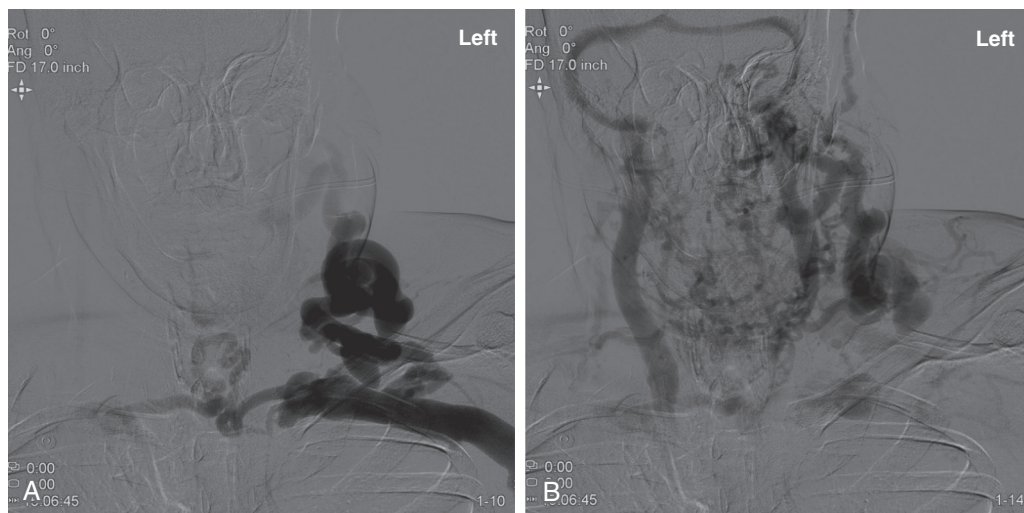


Fig. 68.24 A, Digital subtraction angiography (DSA) showing a complete occlusion of the left innominate vein with severe enlargement of the left internal jugular and multiple ipsilateral and across the neck collaterals draining into a normal right innominate vein. B, DSA of the same patient showing the drainage of the left internal jugular to the right internal jugular across the cavernous veins.

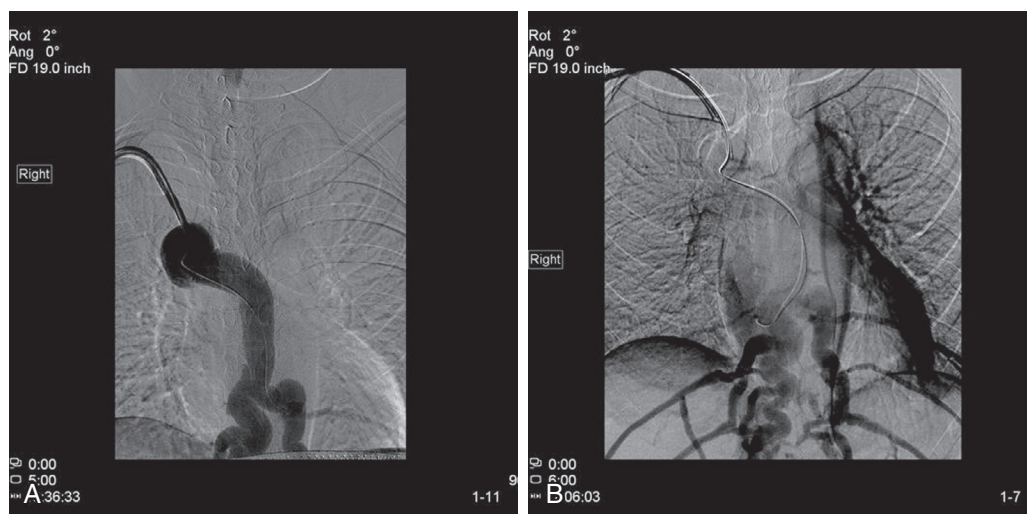


Fig. 68.25 A, Digital subtraction angiography (DSA) showing a complete occlusion of the superior vena cava with severe augmentation of the Azygos vein. B, DSA of the same patient showing the drainage of the azygos vein into the lumbar and intercostal vein.

stenosis, as compared to stenoses at more peripheral locations.⁸⁴ As a result, patients with central vein stenosis may require multiple angioplasties to treat recurrent lesions.

Stent placement has been attempted in the management of refractory central vein stenosis due to elastic recoil (Fig. 68.26A, 68.26B, 68.26C). Several small series have reported the outcomes of stent placement for refractory central venous stenotic lesions. These studies have been limited by their retrospective study design, the small numbers of patients, and the absence of a control group. In two uncontrolled series, the primary patency following stent deployment for central vein stenosis was 42% to 50% at 6 months, and only 14% to 17% at 1 year.^{198,201} Although there are no published randomized studies comparing stent deployment with angioplasty of central vein stenosis, the primary patency utilizing stents appears no better than that achieved with angioplasty alone.²⁰⁶ In patients with ipsilateral vascular access and persistent upper extremity edema despite attempted angioplasty, the only recourse may be ligation of the vascular access, creation or placement of a contralateral arteriovenous access, transition to peritoneal dialysis, or urgent kidney transplantation.

ACCESS-INDUCED HAND ISCHEMIA

The clinical manifestations of access-induced hand ischemia can vary widely, ranging from mild (hand pain) to severe (tissue necrosis with loss of digits). Vascular accesses have a normal retrograde blood flow, which does not cause hand ischemia. There are three recognized mechanisms for access-induced hand ischemia: (1) True arterial steal (with complete retrograde blood flow toward the graft or fistula); (2) Arterial stenosis distal to the arterial anastomosis, and (3) Generalized arterial calcification (most commonly seen in patients with diabetes mellitus) with distal digital occlusions (Fig. 68.27A, 68.27B, 68.27C). Since all three mechanisms may coexist in a single patient, “distal hypoperfusion ischemic syndrome” is the preferred term.²⁰⁷ Other causes of hand pain, such as carpal tunnel syndrome, arthropathies, diabetic neuropathy,

ischemic monomelic neuropathy and reflex sympathetic dystrophy syndrome, should be excluded before making a definitive diagnosis of vascular access-induced hand ischemia.

Physical examination is the most important tool to diagnose access-induced hand ischemia. Pale, cold digits during and between hemodialysis sessions are seen in almost 90% of the cases. There are several additional diagnostic tools including: digital brachial index measurement, digital plethysmography, duplex ultrasound, and transcutaneous oxygen saturation. If these noninvasive tests are suggestive of access-induced hand ischemia, then a complete angiogram of the vascular access, including the feeding artery, should be performed.

Management of access-induced hand ischemia is directed at alleviating the underlying etiology. Treatment options range from angioplasty of the stenosis distal to the anastomosis to ligation of the vascular access. Other interventions that have been successful in relieving the ischemia while maintaining patency of the AV fistula include banding or plication of the AV fistula, graft interposition, and distal revascularization with interval ligation (DRIL).²⁰⁸

INDWELLING HEMODIALYSIS CATHETERS

NON-TUNNELED TEMPORARY HEMODIALYSIS CATHETERS

Temporary hemodialysis catheters are indicated for acute dialysis treatments. They are made of polyurethane, polyethylene, or polytetrafluoroethylene; they have a double lumen, are semi-rigid, and relatively easy to place in the internal jugular (preferably on the right side), femoral, and rarely subclavian veins. Each site has its advantages and disadvantages, but if they are placed in the femoral vein, the catheter should not stay longer than 72 hours, and the internal jugular vein catheters not longer than 1 week, due to the high risk of bacteremia with longer dwell times.²⁰⁹ The subclavian vein is usually used only if there is no other access available since it is more difficult and a higher risk puncture (pneumothorax) and there is an increased risk for central venous stenosis

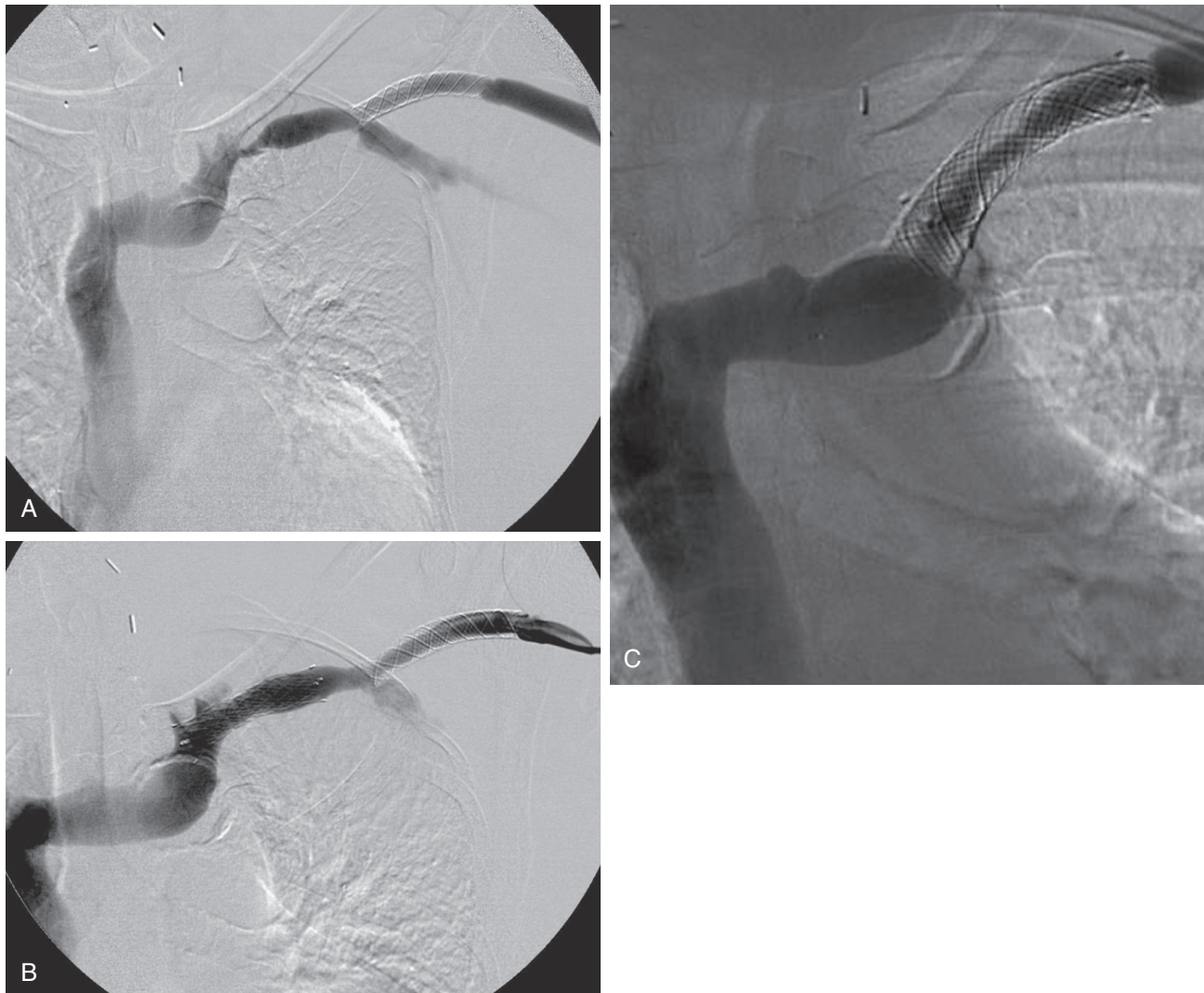


Fig. 68.26 A, Digital subtraction angiography (DSA) showing a severe stenotic lesion of the left subclavian. There is also a stent in the left cephalic vein. B, DSA showing a stent deployed at the severe stenotic lesion of the left subclavian with excellent initial results. C, DSA taken 12 months after the initial placement of a stent in the left subclavian showing intra-stent stenosis due to significant myointimal hyperplasia.

with occlusion of the subclavian and central vessels, which can compromise creation or placement of future AV fistulas or grafts on the same side.¹⁹⁶ If the upper vessels are used, the catheter should be long enough to have its tip at the junction of the right atrium and superior vena cava; if the femoral vein is used, the catheter's tip should be located in the inferior vena cava. If the patient is expected to remain catheter-dependent for a longer time period, a tunneled catheter should be placed. Temporary hemodialysis catheters can be placed blindly, by ultrasound guidance or by fluoroscopic guidance. A chest radiograph should always be obtained immediately following placement of a central vein dialysis catheter before hemodialysis is initiated to confirm line placement and exclude pneumothorax; it is not needed after placement of a femoral dialysis catheter.

TECHNICAL PROCEDURE: INSERTION OF TEMPORARY DIALYSIS CATHETERS

The procedure is usually performed at the patient's bedside, but occasionally performed in the interventional suite. The

patient must be fully informed of the risks and benefits of the procedure. Strict sterile technique and use of local anesthesia is indicated. Access to the femoral or internal jugular vein may be done either blindly or by real-time ultrasound guidance. Real-time ultrasound is highly recommended, as it decreases the number of attempts at vein cannulation, and minimizes the risk of inadvertent arterial cannulation.

Potential complications at time of placement at the upper vessels include: pneumothorax, vein or arterial perforation, mediastinal or pericardial perforation with possibility of hemothorax and cardiac tamponade, air embolism, and local hematoma with possible extension into the soft subcutaneous tissue of the neck and possible external obstruction of the airways. The long-term complications include development of stenotic lesions along the trajectory of the catheter, which may preclude the use of the ipsilateral limb for future creation of a vascular access. If the patient already has a documented stenotic lesion of the central vessels, placement of an indwelling catheter may cause life-threatening acute central vessel

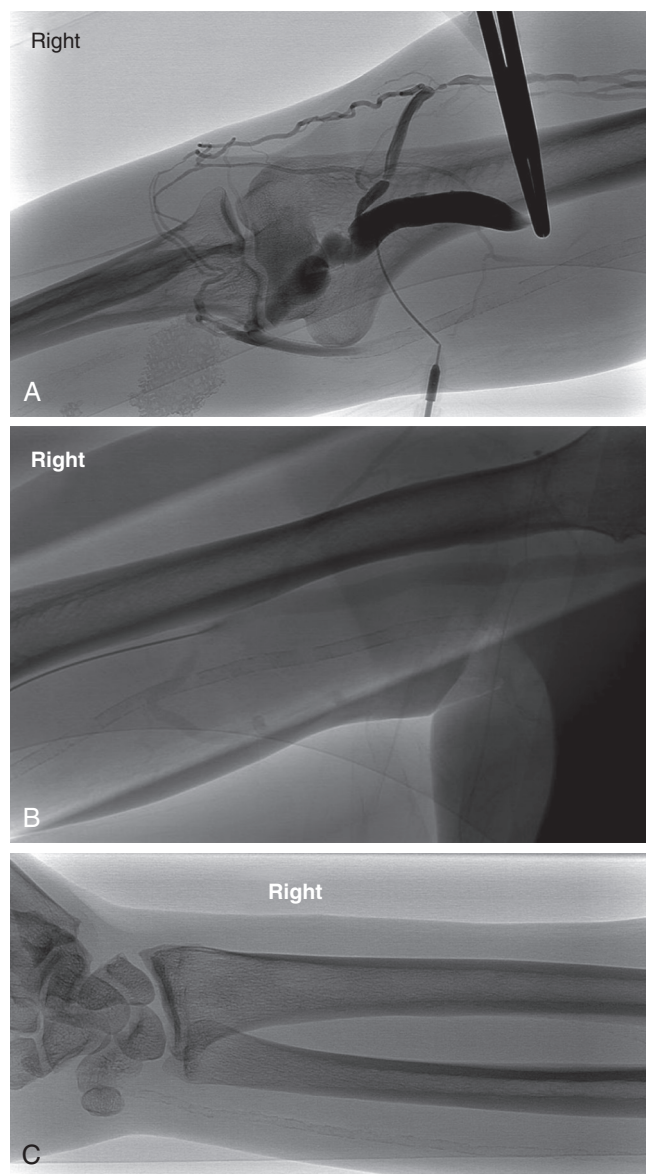


Fig. 68.27 A, Angiogram of a right upper arm fistula demonstrating an arterial steal syndrome on a patient with generalized arterial disease. B, Spot film of the same patient showing a calcified braquial artery. C, Spot film of the same patient showing a calcified ulnar artery.

occlusion. Exit-site infections and catheter-related bacteremia are frequent complications of temporary dialysis catheters. Development of catheter-related bacteremia requires institution of systemic antibiotics and removal of the non-tunneled dialysis catheter.

The complications at the femoral site are less dramatic, but vein or arterial perforation and formation of arteriovenous fistula are possible. Deep vein thrombosis, (either related to the catheter itself, or immobilization related to prescribed bed rest) local hematomas, exit-site infections, and bacteremia are other important complications.

TUNNELED HEMODIALYSIS CATHETERS

Tunneled hemodialysis catheters are also commonly used for vascular access in patients waiting for placement or



Fig. 68.28 Access to the right internal jugular vein guided by real-time ultrasound.

maturation of a permanent vascular (AV fistula or graft). They are also required for long-term access in patients who have exhausted all options for placement of a permanent AV access in all four extremities. Tunneled dialysis catheters are usually placed in a central vein in the chest, most commonly through the internal jugular vein, and rarely in the subclavian vein. They have the same characteristics as temporary catheters, but are longer and have a Dacron cuff located in the tunneled portion of the catheter in the subcutaneous tissue. An inflammatory response around the cuff results in scar tissue, creating a mechanical barrier that prevents introduction of infection from the exit site into the bloodstream. As a result, the frequency of catheter-related bacteremia is lower with tunneled dialysis catheters as compared with acute non-tunneled catheters.^{210,211}

TECHNICAL PROCEDURE: INSERTION OF TUNNELED HEMODIALYSIS CATHETERS

Strict sterile technique, topical local anesthesia (1% Xylocaine), and frequently conscious sedation are used. Access to the internal jugular vein is guided by real-time ultrasound (Fig. 68.28). A 21-gauge needle is used for access. A wire is passed down into the IVC under fluoroscopy guidance. A skin pocket of about 1 cm is created at this location. The permanent indwelling hemodialysis catheter is attached to a tunneler device and a tunnel is created lateral, inferior, under the clavicle, approximately 5 to 7 cm from the initial needle insertion. (Fig. 68.29). The tip of the catheter is introduced and fed up to the junction of the superior vena cava and right atrium (Fig. 68.30). The cuff should be 2 cm from the exit site. A final X-ray is taken to assess for kinks of the catheter and for placement (Fig. 68.31). The catheters are 14.5 or 16 French with lengths from 24 cm for right internal jugular, 28 cm for left internal jugular, and from 36 to 42 cm for femoral veins.

Complications at the time of placement are similar to those associated with temporary catheters (Figs. 68.32, 68.33, 68.34, 68.35). Internal jugular thrombosis develops in about 25% of tunneled catheters, but is usually asymptomatic.²¹² Other long-term complications include dysfunction due to intraluminal thrombosis or fibrin sheaths, exit-site infections, tunnel infections, and catheter-related bacteremia.⁶

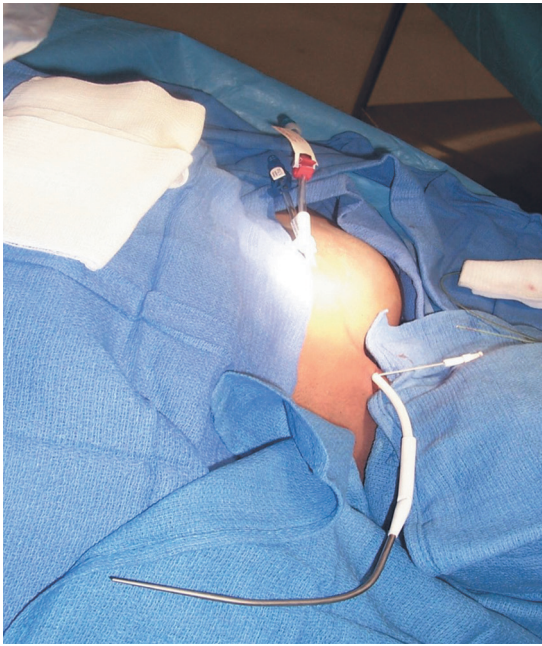


Fig. 68.29 The catheter has been tunneled from the upper part of the chest to the internal jugular site entrance.



Fig. 68.31 Spot film of an appropriate placement of a right internal jugular tunneled chronic dialysis catheter.

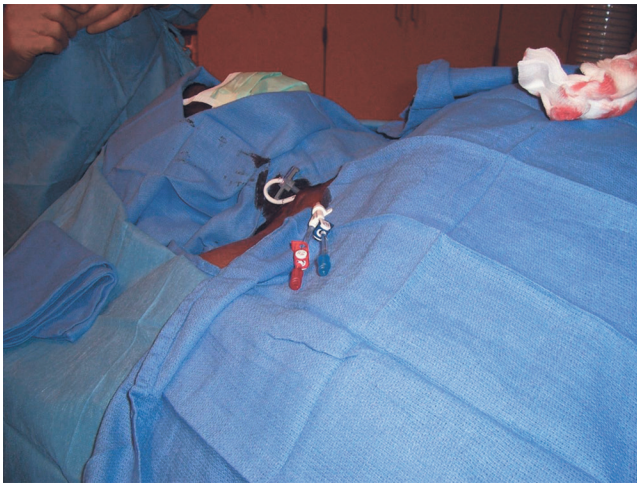


Fig. 68.30 The catheter is introduced into the opening of the peel-away sheath and fed up to the junction of the superior vena cava and right atrium

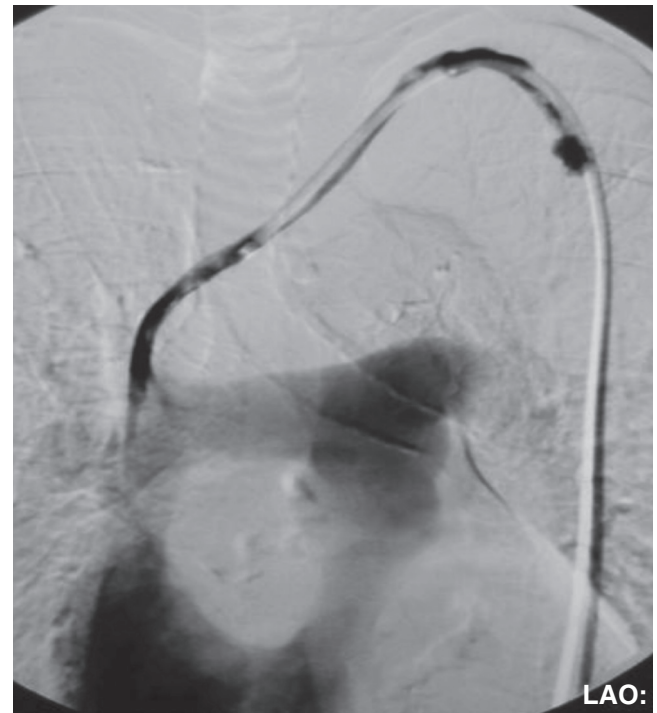


Fig. 68.32 Tunneled catheter complication: Fibrin sheath around the left internal jugular vein catheter.

Tunneled dialysis catheters can be removed at bedside.²¹³ Fulop et al. have recently published a bedside procedure of tunneled dialysis catheter removal with safe and successful outcomes, providing a detailed description of supplies and procedure technique.²¹³ In brief, the precise location of the “cuff” is identified followed by cleansing of catheter exit area and the preparation of the sterile field.²¹³ Next, the subcutaneous tissue surrounding the dialysis catheter up to the cuff is bluntly dissected with a hemostat clamp in order to achieve mobilization of the dialysis catheter from the surrounding tissue.²¹³ Once the cuff and dialysis catheter are free from the surrounding connective tissue, the dialysis catheter should be pulled with a measured level of force.²¹³ Hemostasis should be achieved by applying direct pressure over the proximal

venous puncture site (and not only the exit site) with a gauze for 10 to 15 minutes until bleeding is not detectable.²¹³

LESS COMMON LOCATIONS FOR TUNNELED HEMODIALYSIS CATHETERS

If prolonged use of upper extremity dialysis catheters leads to bilateral central vein occlusion, it becomes necessary to place



Fig. 68.33 Tunneled catheter complication: The catheter was placed in the right carotid artery with the tip near the aortic valves. Contrast demonstrates the aortic arch.

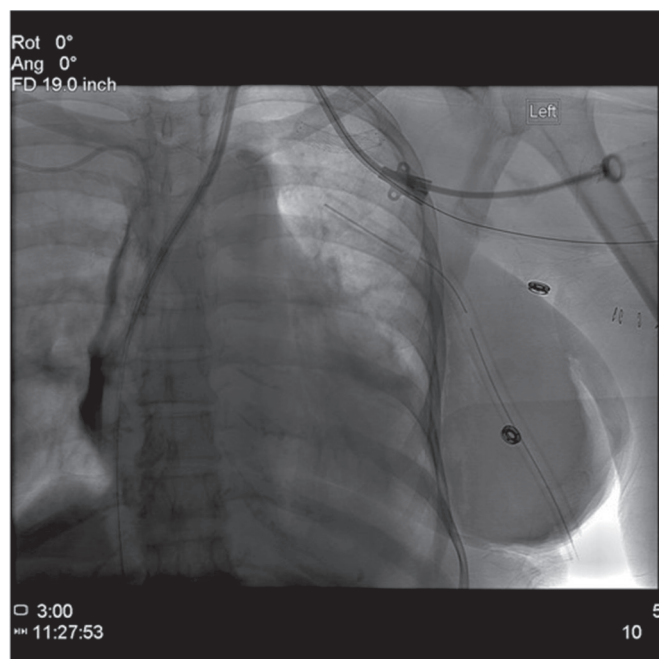


Fig. 68.34 Tunneled catheter complication: Left internal jugular vein catheter with its tip located in the pericardial sac. Contrast demonstrates the outer lining filling defect that corresponds to the lower part of the adventitia of the superior vena cava and upper part of the pericardium.

a *tunneled catheter in the femoral vein*.^{214,215} The procedure for placement of a tunneled femoral catheter is similar to that for a tunneled internal jugular vein catheter, except that a longer (36 to 42 cm) catheter is required, and the catheter tip is placed in either the proximal inferior vena cava or in the right atrium (Fig. 68.36A and 68.36B).²¹⁴ The subcutaneous tunnel is created in the anterior upper thigh. The primary patency of tunneled femoral catheters is significantly worse than that of tunneled internal jugular catheters.²¹⁴ Presumably, some failures are due to kinking of the catheter in the groin when the thigh is flexed. The frequency of catheter-related

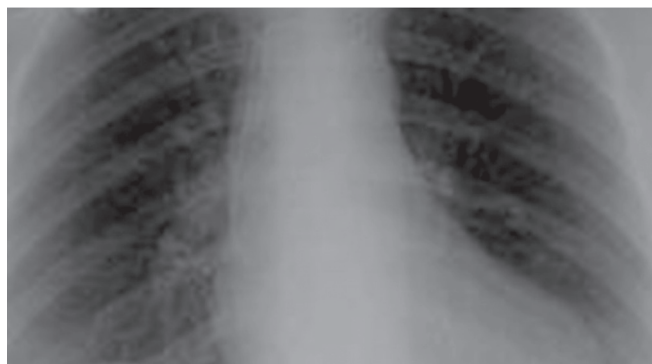


Fig. 68.35 Tunneled catheter complication: Foreign body (metallic wire) that was left behind after an attempt for an internal jugular vein catheter placement.

bacteremia is similar for patients with femoral and internal jugular dialysis catheters. The likelihood of catheter-related bacteremia is proportional to the duration of catheter use.^{197,216} There is high (~25%) frequency of symptomatic ipsilateral deep vein thrombosis after placement of a tunneled femoral catheter.²¹⁴ Fortunately, this complication can be treated with long-term anticoagulation, thereby permitting continued use of the catheter. In patients on hemodialysis in whom the central veins in the chest and groin have been exhausted, the placement of tunneled dialysis catheters at unconventional sites (translumbar and transhepatic) has been described. Catheters at these locations should be considered as last-resort options, as they are associated with a substantial risk for complications.

For *translumbar catheters* there are essentially two tunnels. One tunnel extends from the access site in the back toward the IVC and the second from the abdomen to the access site in the back. The patient is placed on the angiography table in a lateral position with his or her left side down. The initial access site is in the right lower back postero-laterally just above the iliac wing. The needle is directed toward the inferior vena cava under fluoroscopy guidance; once venous access is achieved a glide-wire is placed. A tunnel is created from a lower abdominal site approximately 10 cm from the initial needle access around the waist toward the initial needle access. The permanent hemodialysis catheter is advanced through the tunnel and the cuff is buried into the adjacent subcutaneous tissue (Fig. 68.37).^{217,218} Dilation followed by peel-away sheath insertion is done from the back access to the IVC. The dialysis catheter is then inserted through the sheath to the IVC, preferably over the glide wire, which is passed in a retrograde fashion all the way through the catheter. The access site is sutured and catheter is secured in place.

Placement of tunneled trans-lumbar hemodialysis catheters requires greater expertise in interventional techniques than jugular or femoral access especially since ultrasound guidance is not possible. The risk of bleeding and retroperitoneal hematoma is considerably higher than that associated with tunneled femoral vein catheters. The most common complication of translumbar catheters in one series of 10 patients is partial dislodgment of the catheters.²¹⁸

Interventional radiologists at some centers have placed *transhepatic catheters*. The right upper quadrant is prepped

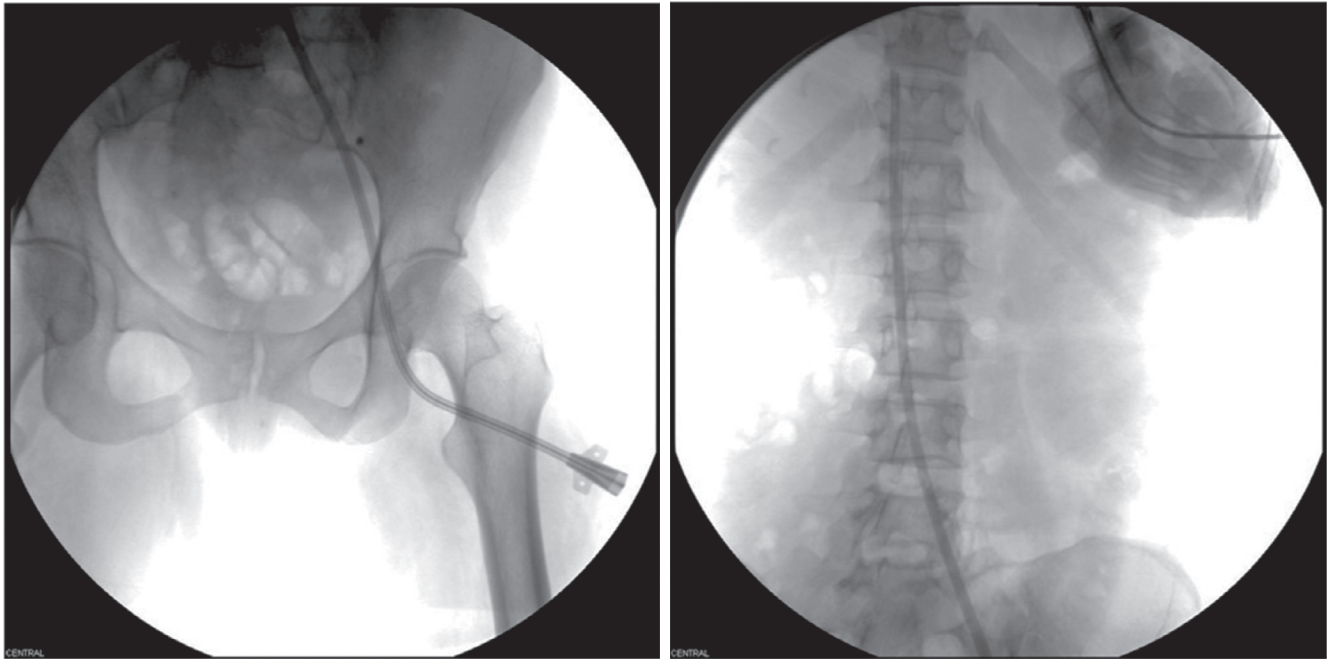


Fig. 68.36 Spot films of an appropriate placement of a left femoral vein tunneled chronic dialysis catheter.

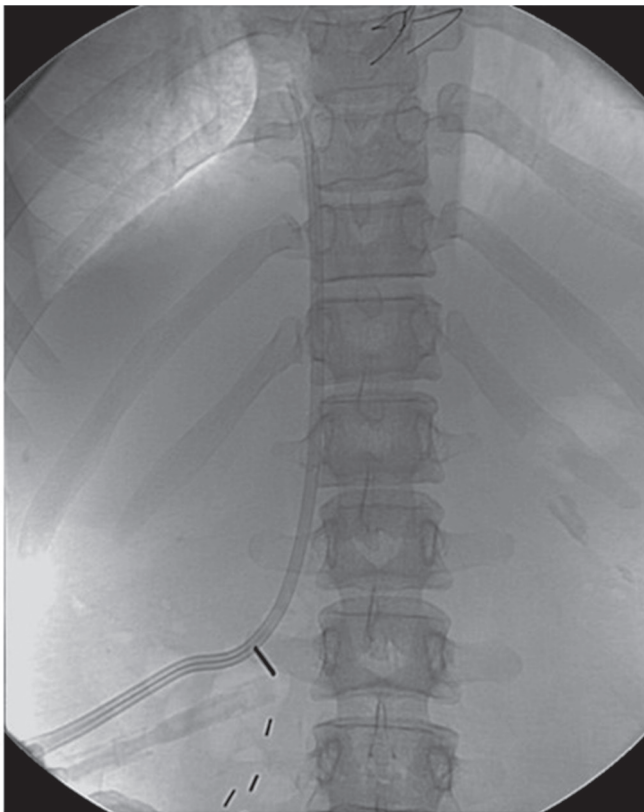


Fig. 68.37 Spot film of an appropriate placement of a translumbar tunneled chronic dialysis catheter.

and draped in the usual manner; a 21-gauge needle is placed halfway through the liver in a direction parallel to the right of the middle hepatic veins under fluoroscopy guidance, contrast material is injected through the needle and the needle withdrawn until a hepatic vein is visualized. Once a

suitable vein is accessed, a guidewire is placed and advanced to the right atrium. A subcutaneous tunnel is created inferiorly to the insertion site and a dual lumen cuffed hemodialysis catheter is placed.^{219–221} The major complications are bleeding and peri-hepatic hematoma. Hepatic vein thrombosis is likely, and dislodgment of catheter due to liver excursion movement with breathing is frequent. Stavropoulos et al. reported a series of 36 transhepatic dialysis catheters placed in 12 patients.²²⁰ The mean survival of these catheters was only 24 days. The thrombosis rate was 2.40 per 100 catheter-days, with the poor catheter patency rates being associated with a high rate of late thrombosis.

EXCHANGE OF TUNNELED HEMODIALYSIS CATHETERS

There are two major indications for catheter exchange: dysfunction and infection. Catheter dysfunction is diagnosed when blood cannot be aspirated from the catheter lumen at the time of dialysis initiation, or more commonly, if it is not possible to consistently achieve a blood flow required to yield sufficient urea and other solute clearance (>250 mL/min). In catheters that were previously delivering an adequate blood flow intraluminal thrombus is the most common etiology for dysfunction, although a fibrin sheath may be the culprit in some patients. This problem is usually treated empirically in the dialysis unit by instilling t-PA into the catheter lumens.²²² Tissue plasminogen activator (tPA) instillation is successful in about 70% to 80% of catheters, but problems with poor flow frequently recur within 2 to 3 weeks. If the thrombolytic agent does not improve the catheter flow, the patient is referred for catheter exchange.

An exit-site infection alone usually resolves with topical antimicrobial agents or oral antibiotics, and is not usually an indication for catheter removal. However, if the patient has a tunnel track infection, catheter removal is mandatory. Catheter-related bacteremia (CRB) is a common indication

for catheter replacement.⁵ In one series, the cumulative risk of CRB among catheter-dependent patients was 35% at 3 months and 48% at 6 months.²¹⁶ Similar risks were observed in a larger prospective study of incident tunneled dialysis catheters.¹⁹⁷ Catheter-related bacteremia is suspected when a catheter-dependent patient experiences fever or chills, and is confirmed by blood cultures from the catheter and a peripheral vein (or the dialysis bloodline) growing the same organism.⁵ When a single set of blood cultures is positive, CRB is still the most likely diagnosis, in the absence of clinical evidence of an alternative source of infection.

The clinical management of CRB has evolved in the past few years.⁵ In the subset of patients whose fever persists after 48 to 72 hours despite appropriate systemic antibiotics (~10% to 15% of patients with CRB), removal of the infected catheter is mandatory. For the remaining patients, there are several management options. The first option is to continue systemic antibiotics without removal of the infected catheter. Unfortunately, infection is infrequently eradicated with this approach; once the course of systemic antibiotics has been completed, bacteremia recurs in 63% to 78% of patients.^{223–227} Moreover, delays in removing an infected catheter may result in metastatic infection, such as endocarditis, septic arthritis, or epidural abscess.²²⁸ The second option is therefore prompt removal of the catheter which removes the source of infection, although this necessitates placement of a temporary (non-tunneled) dialysis catheter, if possible a day or two later to avoid immediate contamination of the new catheter. Once the bacteremia has resolved, a new, tunneled catheter is placed. In an effort to reduce the number of required access procedures, a number of investigators have evaluated the strategy of exchanging the infected catheter for a new one over a guidewire. Several publications have documented the safety and efficacy of this approach.^{226,229–232}

In the past few years, there has been a growing recognition of the central role of bacterial biofilms in causing CRB. Biofilms develop on the inner surface of the lumens of central vein catheters in as little as 24 hours, and are relatively refractory to conventional plasma concentrations of antibiotics.^{233–235} The third option is instillation of a concentrated antibiotic solution into the catheter lumen after each dialysis session (“antibiotic lock”) which can frequently kill the bacteria in the biofilm. This approach can potentially remove the source of the infection (the biofilm), while permitting catheter salvage. The use of antibiotic locks, in conjunction with systemic antibiotics, has been shown to eradicate infection, while salvaging the catheter, in about two thirds of the patients.^{5,236–239} This strategy is not associated with an increased risk of metastatic infections, as compared to prompt catheter removal or exchange of the infected catheter over a guidewire. In our experience, implementation of an antibiotic lock protocol has dramatically reduced the frequency of catheter exchanges due to infection. The overall success rate with the addition of an antibiotic lock in conjunction with systemic antibiotics in curing an episode of catheter-related bacteremia is about 70%.²⁴⁰ However, the success rate varies considerably depending on the bacterial organism, being highest for gram-negative infections, intermediate for *Staphylococcus epidermidis* and *Enterococcus* infections, and lowest for *Staphylococcus aureus* infections.^{238,239,241–243} Catheter-related candidemia always requires catheter exchange, in conjunction with antifungal therapy.²⁴⁴ If fever and/or bacteremia persist

despite a trial of an antibiotic lock, the catheter should be removed or exchanged over a guidewire.⁵

Thrombus inside the catheter may act as a nidus for the catheter biofilm. Heparin coating of the catheters may prevent bacterial adherence.²⁴⁵ Two randomized clinical trials with short-term, non-tunneled central vein catheters found a lower rate of catheter-related bacteremia in patients with heparin-coated catheters than those with non-coated catheters.^{245,246} A recent retrospective study observed fewer infections in patients with heparin-coated tunneled dialysis catheters, as compared with those using an uncoated catheter.²⁴⁷ A randomized clinical trial is needed to confirm this benefit. A recent study by Hemmelgarn et al. reported that instillation of rt-PA instead of heparin into the dialysis catheter lumen once weekly, as compared with the standard use of heparin locks three times a week, significantly reduced the incidence of catheter malfunction and bacteremia,²⁴⁸ suggesting that rt-PA may prevent thrombosis as well as biofilm development.

PERITONEAL DIALYSIS CATHETER PROCEDURES

Peritoneal dialysis (PD) is an alternative to hemodialysis in patients with ESKD. Although it is widely used in many countries, fewer than 10% of the US dialysis population is treated by this modality.²⁴⁹ Peritoneal catheters can be placed into the abdominal cavity by surgeons,^{250–252} interventional radiologists,²⁵³ or interventional nephrologists.²⁵⁴ There are several techniques: blind (Seldinger) technique,²⁵⁵ surgical placement,²⁵⁰ peritoneoscopic,²⁵⁶ laparoscopic,²⁵⁷ Moncrief-Popovich technique,²⁵⁸ and fluoroscopic insertion.²⁵⁴ Incorporation of PD catheter placement in an established interventional nephrology program increases the utilization of this dialysis modality.²⁵⁹ A modified fluoroscopic technique, which adds real-time ultrasound visualization, avoids the risk of inadvertent epigastric artery injury.²⁶⁰

Peritoneal catheters are made of either silicone rubber or polyurethane. The Tenckhoff catheter is still the most common type of PD catheter placed. The intraperitoneal portion of the catheter can be either straight, coiled, Ash (T-Fluted), or with a silicone disc.²⁶¹ The extraperitoneal portion of the catheter may be straight or have a swan-neck design with single or double inner cuffs, or a combination of a single cuff and a silicone disc. The most widely used PD catheter is the double cuff, swan neck, coiled Tenckhoff design. This design has been shown to decrease mechanical complications, i.e., inflow and outflow problems. The coiled catheter design also decreases pain during infusion, and is less likely to migrate. The swan-neck design was introduced to avoid cuff extrusions.²⁶² The intraperitoneal portion of the catheter should be placed between the visceral and parietal peritoneum near the pouch of Douglas. The inner cuff should be inserted in the abdominal wall musculature (rectus muscle) to prevent leaks. The outer cuff should be located in the subcutaneous tissue to create a dead space in between the two cuffs, which it is believed would prevent migration of infections coming from the exit site. The subcutaneous tract and exit site should face downward and laterally to avoid exit-site infection. The exit site should be determined and marked prior to the insertion while the patient is in the upright position. The beltline, prior surgical sites and the

abdominal midline should be avoided. Postoperative catheter care is very important. The catheter should be covered with a nonocclusive dressing, and should not to be used for 10 to 14 days. The catheter should be flushed at least two or three times per week with saline or dialysate solution until the patient is ready to start PD.²⁶³ Usually PD is started between 2 to 4 weeks after placement of the catheter, to allow for wound healing, and securing of the catheter cuff. Providing sufficient time for healing helps to avoid leaks, which can increase the risk of infection and are discouraging to patients. Low-volume PD may be attempted within 24 hours of catheter placement, if no other dialysis access is available and an urgent start is required.²⁶⁴

Two studies comparing swan-neck and straight Tenckhoff catheters have shown a similar risk for peritonitis and exit infection, but less cuff extrusion with the swan-neck design. The lower incidence of cuff extrusion enhances the survival of the swan-neck catheters.^{265–267} A modified technique, which places the swan-neck catheter in a presternal exit-site location, has been reported by Twardowski et al. and has shown an increase in access survival up to 95% at 2 years. The presternal exit-site modification has been shown to decrease the incidence of peritonitis as well as exit-site infection. The authors advocate the use of the presternal catheters in obese patients, patients with ostomies, and children wearing diapers and/or with fecal incontinence.^{268,269} Gadallah et al. demonstrated that placement of PD catheters by peritoneoscopic approach had a longer survival rate (survival defined as inflow/outflow obstruction, persistent dialysate leak and persistent peritonitis, or exit-site/tunnel infection requiring catheter removal) than those placed surgically, and that the rates of exit infection and leak were lower.²⁷⁰ Moncrief et al. described a technique in which the extraperitoneal portion of the catheter is buried in the abdominal subcutaneous tissue until the patient is ready for PD. It appears that the Montcrief modification may lower the risk of initial infection of the tract.²⁵⁸

A major complication during placement of the PD catheter is bowel perforation. It is infrequent with all techniques except for blind placement, but once identified it requires bowel rest, intravenous (IV) antibiotic therapy, and rarely surgical exploration.^{250,259} Tip migration is a very common (up to 35%) late complication, which could cause problems with draining of the PD fluid. It can be fixed with either radiologic or surgical manipulation.^{271,272} PD leaks around the catheter have been reported as high as 10%.²⁷³ Perioperative infection and bleeding are very rare, as prophylactic antibiotics are usually given.²⁷²

TECHNICAL PROCEDURE: INSERTION OF PERITONEAL CATHETERS BY FLUOROSCOPIC AND ULTRASOUND TECHNIQUE²⁷⁴

The abdomen is prepped and draped in a sterile fashion. Conscious sedation is administered with midazolam hydrochloride and fentanyl citrate in many facilities with these capabilities. A registered nurse obtains vital signs and administers conscious sedation during the procedure. The insertion site is selected to be 2 cm to the left or right and below the umbilicus. An ultrasound machine with a 5 to 12 MHz transducer and a sterile cover is used to guide a 21-gauge needle into the peritoneum. Under ultrasound guidance the needle penetrates through the skin, the subcutaneous tissue, the outer fascia of rectus muscle, the muscle

fibers, the inner fascia, and the parietal layer of peritoneum. Three to 5 mL of radiocontrast are injected into the peritoneal cavity under fluoroscopy to assure appropriate location: a radiologic pattern of outer bowel delineation is indicative of a good placement. A 0.018-inch cope-mandrel-wire is introduced through the needle. The needle is exchanged for a 6-French catheter sheath. A 2-cm incision is made on the skin and the subcutaneous tissue is digitally dissected up to the rectus muscle. A series of dilators (8-, 12-, and 14-French) are passed over a stiff glide-wire, and an 18-French peel-away sheath is placed. A double-cuff, swan-neck, Tenckhoff peritoneal dialysis catheter is introduced over the stiff glide-wire into the peritoneal cavity. The coiled intraperitoneal portion is placed in the lower intraabdominal area. The inner cuff is pushed into the muscle before removing the peel-away sheath. Alternatively, the tunnel can be created before catheter insertion into the peritoneum. A tunnel is created with an exit site located distal, lateral, and below the initial incision with the outer cuff buried in the subcutaneous tissue. A final fluoroscopic imaging is performed to verify placement of the Tenckhoff catheter (Fig. 68.38A, 68.38B, 68.38C). Inflow and outflow of the PD catheter is tested with 500 mL of normal saline. The PD catheter is flushed with 10 to 15 mL of heparin. The subcutaneous tissue and skin are sutured, and the site is dressed.

TECHNICAL PROCEDURE: INSERTION OF PERITONEAL CATHETERS BY PERITONEOSCOPIC TECHNIQUE

Peritoneal catheters placed peritoneoscopically are implanted through the rectus muscle using the Y-TEC technique. It has the same initial preparation as for the fluoroscopic/ultrasound technique. Under local anesthesia, a 2-cm skin incision is made. The subcutaneous tissue is dissected up to the rectus muscle. A catheter guide is inserted into the abdomen and the Y-TEC peritoneoscope into the catheter to assess initial entry to the peritoneal cavity. The scope is removed and 500 mL of air infused into the cavity. The scope is again replaced and advanced to the pelvic area. This area is inspected for adhesions and bowel loops. The scope is again removed and the peritoneal catheter is introduced through the catheter with the help of a stainless steel stylette. The catheter is advanced to the pelvic area. The stylette is removed and the inner cuff is buried into the musculature. The exit location is determined and the catheter is tunneled to that location.

TECHNICAL PROCEDURE: INSERTION OF PERITONEAL CATHETERS BY PRESTERNAL CATHETER PLACEMENT

The peritoneal catheter implantation technique is the same as the peritoneoscopic insertion, except that the PD catheter has a straight design instead of a swan neck. After the PD catheter is placed, then a second catheter is tunneled from the midabdomen up to the chest wall. The two catheters are connected by a titanium joint piece. The second catheter has the swan-neck design and two cuffs. The exit site is located lateral to the mid-sternal line.

PERCUTANEOUS KIDNEY BIOPSY (SEE ALSO CHAPTER 26)

A percutaneous kidney biopsy is an important procedure in the diagnosis of acute and chronic kidney disease. The results

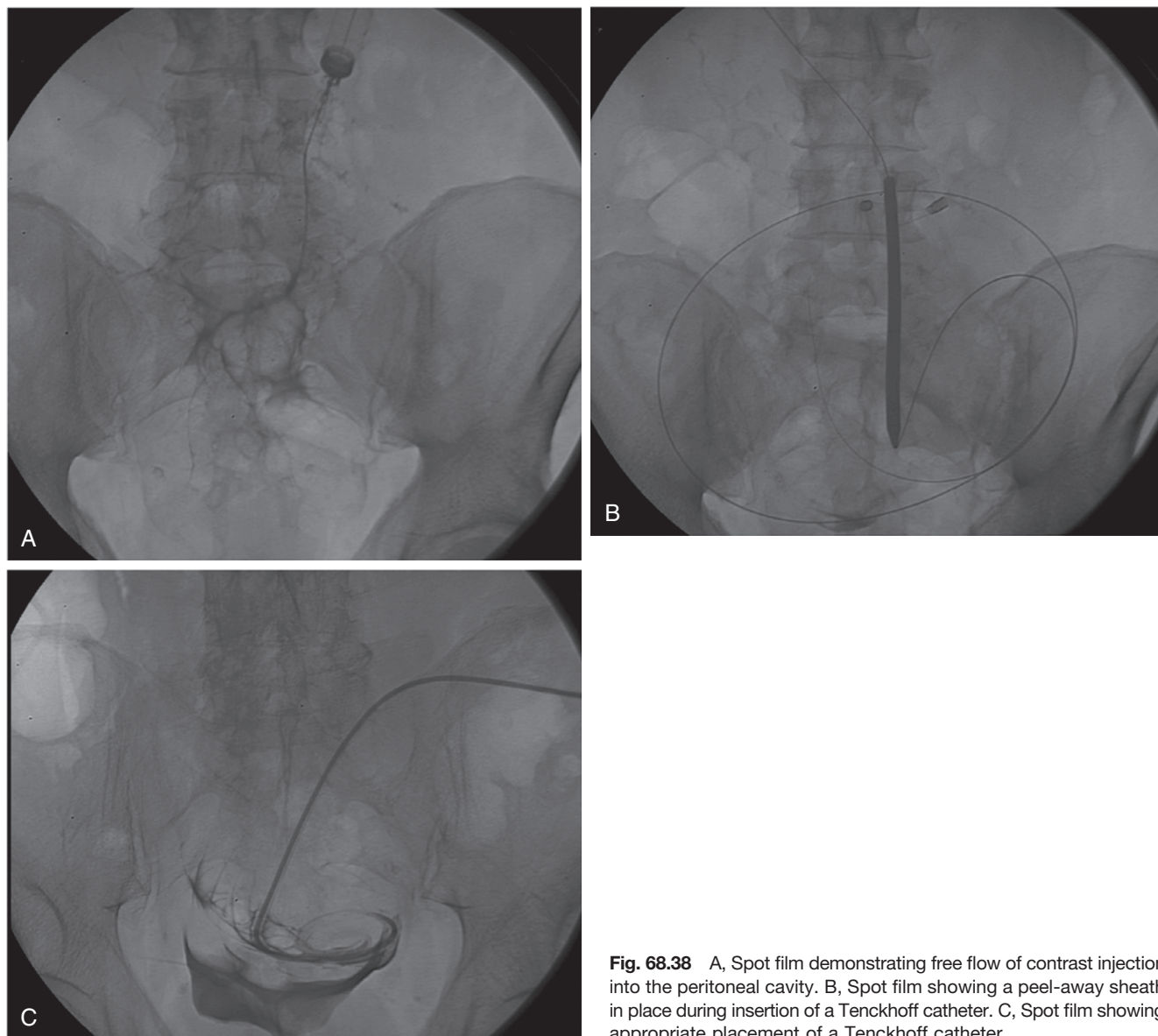


Fig. 68.38 A, Spot film demonstrating free flow of contrast injection into the peritoneal cavity. B, Spot film showing a peel-away sheath in place during insertion of a Tenckhoff catheter. C, Spot film showing appropriate placement of a Tenckhoff catheter.

of a kidney biopsy are helpful in guiding medical therapy and providing a prognosis. The goal of a kidney biopsy should be to maximize the yield of adequate tissue while minimizing the risk of complications. Percutaneous kidney biopsies have evolved from a blind procedure to a real-time ultrasound-guided needle biopsy. At some institutions, radiologists perform kidney biopsies under ultrasound or CT guidance. Although some nephrologists still use the Franklin-Silverman needle and the Tru-cut needle for blind biopsy, several authors have documented that the use of real-time guided ultrasonography along with the use of an automatic biopsy gun minimizes complications and provides a high yield of adequate tissue for pathologic diagnosis (Table 68.13).^{275–282} Cozens et al. retrospectively compared a 15-gauge Tru-cut renal biopsy with ultrasound localization and marking to use of an 18-gauge spring-loaded gun renal biopsy under real-time ultrasound guidance. They reported a 79% yield of adequate renal tissue with the blind technique (15 gauge), as compared with 93%

with real-time ultrasound guidance (18 gauge).²⁸⁰ Similarly, two other studies reported a higher mean number of glomeruli from biopsies obtained under real-time ultrasound, as compared to those performed blindly.^{279,281}

Major complications of kidney biopsies, including gross hematuria or retroperitoneal hematoma requiring blood transfusion, invasive procedure, or surgical intervention have been reported in less than 1% of biopsies in some series, and 5% to 6% in others (Table 68.13). The likelihood of major complications was not associated with patient age, blood pressure, or serum creatinine in one large series.²⁸¹ Among those patients with major complications, the time interval from biopsy to diagnosis of the complication was ≤ 4 hours in 52%, ≤ 8 hours in 79%, and ≤ 12 hours in 100% of patients.²⁸¹ A recent prospective study of 100 consecutive patients showed that outpatient, real-time, ultrasound-guided percutaneous renal biopsy is safe and minimizes the need for post-biopsy hospitalization.²⁸¹ Minor complications, including transient

gross hematuria or perinephric hematoma not requiring transfusion or intervention, occurred after 6.6% of biopsies in one series.²⁸¹ Either ultrasound or computerized tomography can be used to diagnose perinephric hematomas.²⁸³ Most hematomas resolve spontaneously within a few weeks with no significant sequelae. Major bleeding complications that do not resolve with conservative measures require further intervention. In the past, this entailed urgent surgical nephrectomy. However, selective renal arteriogram with embolization of the bleeding arteriole is often able to stop the bleeding in most cases. A review article reported major complications in only 0.3% of cases, with death in fewer than 0.1% among 9595 percutaneous kidney biopsies performed over the last 50 years.²⁸⁴

For patients at high risk of bleeding complications or liver disease with coagulopathy in whom a kidney biopsy is indicated, a transjugular kidney biopsy may be performed by an interventional radiologist or nephrologist. Thompson et al.

reported 91% adequate tissue retrieval with an average of nine glomeruli for light microscopy in 23 patients undergoing transjugular renal biopsy.²⁸⁵ A capsular perforation was encountered in 17 (74%) patients, of whom six (26%) required coil embolization of the bleeding vessel. Two major complications were reported, one arterio-calyceal system fistula and one renal vein thrombosis six days after the biopsy.²⁸⁵ Abbott et al. reported a series of nine patients undergoing transjugular renal biopsy.²⁸⁶ Adequate tissue was obtained from all patients. Capsular perforation occurred in 90% of the patients, and two patients developed gross hematuria requiring transfusion.²⁸⁶ The very high rate of capsular perforation casts real doubts on any advantage of transjugular over percutaneous kidney biopsy.

A bleeding disorder is an absolute contraindication to performing a percutaneous kidney biopsy. However, if it can be corrected medically, and if the potential benefit of doing a biopsy outweighs the potential risk, the biopsy can still be performed. Relative contraindications to kidney biopsy include a solitary kidney, pyelonephritis, perinephric abscess, uncontrolled hypertension, hydronephrosis, polycystic kidney disease, severe anemia, pregnancy, renal masses, and renal artery aneurysms.

TECHNICAL PROCEDURE: PERCUTANEOUS KIDNEY BIOPSY UNDER REAL-TIME ULTRASOUND GUIDANCE

A complete blood count, prothrombin time, and partial thromboplastin time are checked before the procedure, as well if the patient is taking medications such as antiplatelet agents, anticoagulants, and nonsteroidal antiinflammatory agents. The patient is taken to the ultrasound suite and placed in the prone position. An initial ultrasound examination is performed to confirm the presence of two kidneys. Sterile technique is observed and sterile cover placed over the ultrasound probe. The lower pole of the left kidney is preferred for right-handed operators. The skin and subcutaneous tissue are anesthetized with 1% lidocaine. Under real-time ultrasound guidance, a biopsy needle gun is advanced up to the capsule of the kidney (Fig. 68.39A, 68.39B). The patient is asked to hold his breath, and the spring-loaded gun is activated. The gun is retrieved and the specimen placed in a container with media. There are different types of needle biopsy guns: full-core, half-core, or $\frac{3}{4}$ of a core. Sizes vary

Table 68.13 Adequacy of Kidney Tissue Retrieval and Complications by Real-Time Ultrasound-Guided Percutaneous Renal Biopsy

Reference	Number of Biopsies	Adequate Tissue	Major Complications ^a
Dowd, 1991 ²⁷⁵	23	95.5%	<0.5%
Doyle, 1994 ²⁷⁶	86	99%	0.8%
Hergesell, 1998 ²⁷⁷	1090	98.8%	<0.5%
Donovan, 1991 ²⁷⁸	192	97.8%	<1%
Burstein, 1993 ²⁷⁹	200	97.5%	5.6%
Cozens, 1992 ²⁸⁰		93%	N/A
Marwah, 1996 ²⁸¹	394		6.6%
Maya, 2007 ²⁸²	65	100%	0%

^aDefinitions of major complications differed among studies.



Fig. 68.39 A, Operator showing the technique of a real-time ultrasound kidney biopsy. B, Kidney ultrasound image showing a biopsy needle located at the lower pole of the kidney.

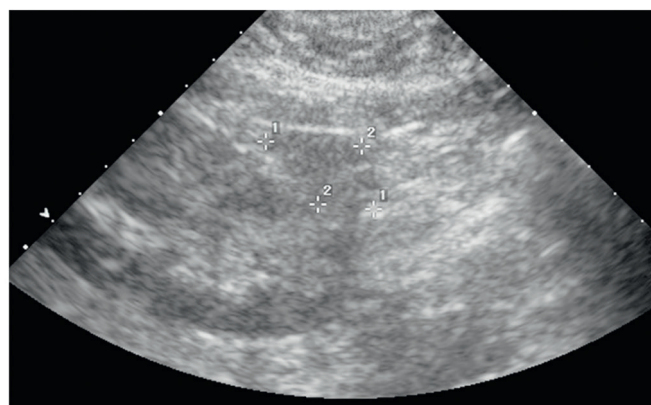


Fig. 68.40 Post-biopsy kidney ultrasound image of a perinephric hematoma. The markers show the size of the perinephric hematoma as a complication of a kidney biopsy.

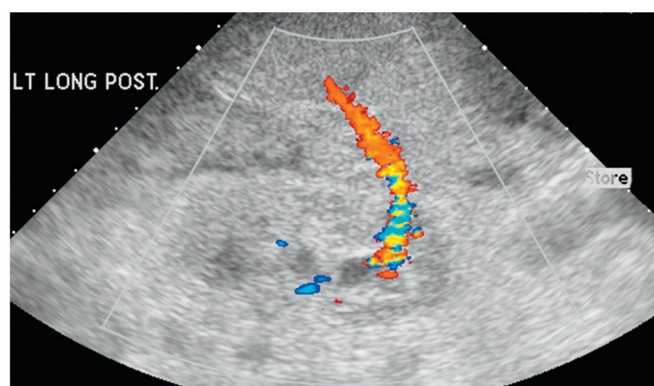


Fig. 68.41 Post-biopsy kidney color Doppler ultrasound image showing active bleeding.

from 14 French to 18 French and lengths from 10 cm to 20 cm. Also, the throw (amount of tissue that the gun can obtain) of the device can be adjusted from 13 mm to 33 mm. Usually two or three biopsy samples are taken in one setting to provide enough tissue for light microscopy, immunofluorescence, and electron microscopy studies. After the biopsy is completed, a second-look ultrasound examination is performed to assess for perinephric hematomas (Fig. 68.40). A color Doppler ultrasound post-biopsy surveillance imaging examination is helpful to localize any active bleeding (Fig. 68.41). The patient remains on bedrest for 4 to 6 hours and vital signs are obtained frequently in the first hour and then hourly over 4 to 6 hours. Hematocrits are checked every 6 hours for the next 24 hours. However, some centers may have protocols for same-day observation and discharge may not need observation for 24 hours.

RADIATION AND PERSONAL SAFETY

The understanding of basic radiation safety is very important to protect the patient, physician, and staff involved in the care of the patient in interventional nephrology. Unnecessary radiation exposure is harmful and can be easily prevented. The Food and Drug Administration (FDA) oversees the rules

and regulations for use of X-ray equipment. The Occupational Safety and Health Administration (OSHA) regulates the radiation exposure of workers. Each state has its own regulatory office to ensure that workers do not exceed a predetermined radiation dose.

Exposure is the amount of ionizing radiation reaching a subject, and is measured in units of Roentgen. The amount of energy absorbed by a material when it is exposed to ionizing radiation is measured in Rads. The absorbed dose is always lower than the exposure, because the tissue does not absorb all the energy from the radiation. An absorbed dose equivalent is used to relate the amount of biological damage and is measured in Rem. The OSHA occupational dose limit for the whole body is 1.25 Rem/quarter; for the extremities it is 18.75 Rem/quarter. A dosimeter must be worn at all times on the outside of the lead apron, and the absorbed dose measured monthly. To protect against radiation the interventionist should minimize the time of exposure to radiation, minimize the use of magnification imaging, properly use collimators and field filters, maximize the distance between the source of radiation and personnel involved with the procedure, minimize the use of cineangiography and continuous fluoroscopy, and use proper shielding, including lead aprons, thyroid collars, leaded glasses, and lead shields.

Knowledge of these facts and application of appropriate safeguards is particularly important in hemodialysis access procedures, especially those involving vascular access interventions in the upper extremity. The operators' proximity to the x-ray tube and difficulty in shielding increases their radiation exposure.

 Complete reference list available at [ExpertConsult.com](https://www.expertconsult.com).

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BOARD REVIEW QUESTIONS

1. What is the most common radiologic lesion present in AVF and AVG dysfunction?
 - a. Arterial stenosis
 - b. Venous stenosis
 - c. Central vein stenosis

Answer: b

Rationale: In several large case series from angiography, venous stenoses were the most common lesion identified in AVF and AVG dysfunction.^{84,158}

2. What are the major endovascular therapies/devices that are *currently available* to “treat” AVF and AVG dysfunction?
 - a. Angioplasty
 - b. Bare metal stents and stent grafts
 - c. Drug-coated balloons
 - d. Drug-eluting stents
 - e. a and b
 - f. a, b, and c
 - g. All the above

Answer: f

Rationale: At present, the major endovascular therapies to treat an AVF and AVG dysfunction include balloon angioplasty, bare metal stents, and stent grafts. There has been recent studies published suggesting the benefit of drug-coated balloons.^{111,112}

3. The major techniques available to assess a failing vascular access include:
 - a. Physical examination
 - b. Angiography
 - c. Ultrasound
 - d. All the above

Answer: d

Rationale: All these techniques are commonly used to assess a failing access. The exact type of technique used may depend on the individual center processes of care.

4. The most common complication(s) resulting from kidney biopsies include:
 - a. Retroperitoneal hematoma
 - b. Gross hematuria
 - c. Hypotension
 - d. Acute kidney injury
 - e. a and b
 - f. None of the above

Answer: e

Rationale: Overall, complications from renal biopsies range from 1% to 6% (see Table 68.13). The most common complications reported are retroperitoneal hematoma and gross hematuria.

5. The major technique(s) to insert a peritoneal dialysis catheter include:
 - a. Blind (Seldinger) technique
 - b. Surgical placement
 - c. Peritoneoscopic placement
 - d. Laparoscopic placement
 - e. Moncrief-Popovich technique
 - f. Fluoroscopic insertion
 - g. a, b, and c
 - h. All the above

Answer: h

Rationale: All the above have been published techniques for peritoneal dialysis catheter placement.^{249–259} Selection of technique may depend on center and operator preference and experience.